

A Unified Formula Atlas for the Thue–Morse Sequence

Digital, valiative, Pascal, automatic, finite-difference, spectral, and analytic representations

Merged and expanded companion report to the Fabius–Rvachev formal development

Vladimir Reshetnikov’s ProveIt repository

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Abstract

Four draft reports in the Fabius-function research-frontier directory developed partly overlapping collections of formulae for the Thue–Morse sequence. This document merges them into one proof-oriented atlas, removes duplicated arguments, reconciles notation and sign conventions, and retains the strongest version of each genuinely different result. The signed sequence $\varepsilon_n = (-1)^{\text{wt}_2(n)}$ and the zero–one sequence $\tau(n) = \text{wt}_2(n) \bmod 2$ are treated simultaneously.

The resulting network begins with binary recursion, block concatenation, XOR characters, floor and fractional-part sums, trigonometric signs, factorial and central-binomial valuations, Catalan valuations, and Kummer carry cocycles. It then develops the Pascal-row viewpoint in several directions: Glaisher’s odd entry count, the existing exact binomial–logarithm formula, a logarithm-free modulo-three formula, sparse power-of-two probes, Boolean Möbius inversion, multinomial support counts, and a sparse Prouhet theorem on the odd support of an arbitrary Pascal row.

The lacunary product

$$\sum_{n \geq 0} \varepsilon_n z^n = \prod_{j \geq 0} (1 - z^{2^j})$$

serves as the central analytic engine. From it we obtain Mahler equations, cyclotomic factorizations, a ruler-function Euler transform, convolution, partition, Bell-polynomial, and Hessenberg-determinant formulae, an algebraic formal series over $\mathbb{F}_2$, an integral algebraic lift, exact finite differences, all dyadic power moments, Walsh and ordinary Fourier transforms, Riesz products, autocorrelation recurrences, iterated prefix sums, and the bridge back to the Fabius and Rvachev functions.

New material added in the unification includes an explicit dyadic weight-distribution chapter, a systematic centered-moment parity theorem, and a Mellin–Dirichlet theory. In particular, the shifted Dirichlet series

$$D(s, a) = \sum_{n \geq 0} \frac{\varepsilon_n}{(n + a)^s}$$

continues to an entire function, satisfies an exact dyadic parameter equation, and vanishes at every nonpositive integer. The proof is driven by the super-polynomial flatness

$$\log \prod_{j \geq 0} (1 - e^{-2^j t}) = -\frac{\log^2(1/t)}{2 \log 2} + \mathcal{O}(\log(1/t)) \quad (t \downarrow 0).$$

A second consolidation pass merged three parallel sibling atlases into this document. It contributed the weighted digit enumerator $\prod_j (1 + uz^{2^j})$ and the subtraction/borrow cocycle; a self-contained proof that the unit circle is a natural boundary, with the Hankel-determinant rigidity theorem and the sharp irrationality exponent $\mu(\Theta_b) = 2$ of the Thue–Morse–Mahler numbers stated from the literature; a Mahler functional equation for the autocorrelation series; a chapter on the combinatorics and dynamics of the infinite word (uniform recurrence, aperiodicity, Thue’s

overlap theorem, critical exponent two, linear factor complexity); and a chapter on rarefied sums – exact residue-class sums modulo three, Gelfond’s optimal exponent $\log_4 3$, a matrix recursion for odd moduli, and Newman’s positivity phenomenon with its modern extensions along primes, squares, and cubes.

Every elementary assertion is proved in the text; deeper historical results about transcendence, substitution spectra, and evaluated products are clearly identified and cited. No mathematical priority is claimed for identities merely derived here, and no formula is silently represented as already formalized in Lean.

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1 Scope, conventions, and the four-draft merge

1.1 Two normalizations

Write the binary expansion of $n \in \mathbb{N} = \{0, 1, 2, \dots\}$ as

$$n = \sum_{j \geq 0} b_j(n) 2^j, \quad b_j(n) \in \{0, 1\}, \quad (1.1)$$

with only finitely many nonzero digits. Its binary weight and the two standard normalizations of the Thue–Morse sequence are

$$\text{wt}_2(n) = \sum_{j \geq 0} b_j(n), \quad \tau(n) = \text{wt}_2(n) \bmod 2 \in \{0, 1\}, \quad \varepsilon_n = (-1)^{\text{wt}_2(n)} = 1 - 2\tau(n). \quad (1.2)$$

Thus

$$\tau(n) = \frac{1 - \varepsilon_n}{2}, \quad \varepsilon_n = 1 - 2\tau(n). \quad (1.3)$$

The signed normalization is usually cleaner for products, transforms, and cancellation; the zero–one normalization is natural for automata, base expansions, and the formula in the source exposition.

We use $a \oplus b$ for bitwise exclusive-or and $k \preceq_2 n$ when every one-bit of k is also a one-bit of n . Empty products are 1, empty sums are 0, and $0^0 = 1$ when it occurs in a finite moment identity. The valuation $\nu_2(m)$ is the exponent of 2 in a positive integer m , with $\nu_2(1) = 0$.

1.2 The Pascal-row formula being complemented

Section 11.8 of the main Fabius–Rvachev exposition defines

$$\mathcal{X}(n) = \sum_{k=0}^n (-1)^{\binom{n}{k}} \quad (1.4)$$

and proves the exact-power identity

$$n + 1 - \mathcal{X}(n) = 2^{\text{wt}_2(n)+1}. \quad (1.5)$$

Consequently

$$\tau(n) = \frac{1 + (-1)^{\log_2 \left(n+1 - \sum_{k=0}^n (-1)^{\binom{n}{k}} \right)}}{2}. \quad (1.6)$$

The logarithm is exact because its argument was first proved to be a power of two. The formula hides the binary expansion inside the parity pattern of a whole Pascal row. The present atlas asks what other structures carry the same parity character.

1.3 Merge policy

The four input drafts were not independent papers: large parts shared the same binary, valuation, generating-product, Prouhet, and Fourier skeleton. A literal concatenation would repeat the same proof several times while obscuring the interesting differences. The unified text therefore follows three rules.

1. A theorem common to several drafts appears once, in the strongest useful form, with one proof and cross-references to its later specializations.

2. Material unique to a draft is retained when it is mathematically distinct: in particular the modulo-three Pascal formula, multinomial logarithms, sparse Pascal Prouhet identities, Boolean Möbius inversion, determinant formula, integer algebraic lift, odious/evil enumerators, autocorrelation, Dirichlet integrals, and the explicit Fabius-prefix bridge.
3. Statements added during the merge are marked by context rather than by novelty rhetoric. The report claims derivation, not historical priority.

This unified atlas was itself produced in parallel: three sibling consolidations of the same four drafts were written independently, and a second merge pass (26 August 2026) folded their distinctive material into the present document under the same three rules. The additions are listed in the abstract and in the concordance of [section C](#).

Proof-status convention. The repository already contains formal developments of the binary recurrence, finite generating product, Prouhet cancellations, iterated prefixes, and the binomial–logarithm evaluator. The paper proofs below make many additional results natural Lean targets, but do not assert that those declarations already exist. A dependency-aware formalization map is given in [section 19](#). Since 26 August 2026 the formalization of this atlas is underway in the same repository: seven new compiled Lean modules cover the Boolean-cube master identities, the valuation and digit chapters, the enumerators, the general finite-difference engine, the sparse Prouhet identities, the modulo-three rarefaction, and aperiodicity. Statements proved in Lean carry their declaration names in the text, typeset as `Fabius.thueMorseSign`; see [section 19.1](#) for the module map. Several are proved in Lean in **more** generality than stated here, and the text records the stronger forms.

1.4 A compact formula index

The table is a navigation device rather than a substitute for hypotheses and proofs. Here $P_m(z) = \sum_{n < 2^m} \varepsilon_n z^n$, and $E(z) = \sum_{n \geq 0} \varepsilon_n z^n$.

Viewpoint	Representative exact formula
Binary recursion	$\varepsilon_{2n} = \varepsilon_n, \varepsilon_{2n+1} = -\varepsilon_n.$
Block concatenation	$\varepsilon_{2^m q + r} = \varepsilon_q \varepsilon_r$ for $0 \leq r < 2^m.$
XOR character	$\varepsilon_{a \oplus b} = \varepsilon_a \varepsilon_b.$
Digit product	$\varepsilon_n = \prod_j (1 - 2b_j(n)).$
Floors	$\varepsilon_n = (-1)^{\sum_{j \geq 0} \lfloor n/2^j \rfloor}.$
Fractional parts	$\text{wt}_2(n) = \sum_{j \geq 1} \{n/2^j\}.$
Factorials	$\varepsilon_n = (-1)^{n - \nu_2(n!)}.$
Central binomial	$\varepsilon_n = (-1)^{\nu_2 \binom{2n}{n}}.$
Carries	$\varepsilon_{a+b} = \varepsilon_a \varepsilon_b (-1)^{\nu_2 \binom{a+b}{a}}.$
Pascal support	$\#\{k : \binom{n}{k} \text{ odd}\} = 2^{\text{wt}_2(n)}.$
Sparse Pascal probes	$\varepsilon_n = (-1)^{\sum_j \binom{n}{2^j}}.$
Log-free Pascal residue	$\tau(n) = (2^{\text{wt}_2(n)} \bmod 3) - 1.$
Multinomial support	$O_r(n) = r^{\text{wt}_2(n)}.$
Boolean Möbius	$\mu_2(k, n) = \varepsilon_{n \oplus k}$ for $k \preceq_2 n.$
Finite product	$P_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}).$
Infinite product	$E(z) = \prod_{j \geq 0} (1 - z^{2^j}), z < 1.$
Mahler equation	$E(z) = (1 - z)E(z^2).$

Viewpoint	Representative exact formula
Euler transform	$\log E(z) = -\sum_{r \geq 1} (2^{\nu_2(r)+1} - 1) z^r / r.$
Finite differences	$\sum_{n < 2^m} \varepsilon_n f(x+n) = (-1)^m \Delta_1 \cdots \Delta_{2^{m-1}} f(x).$
Prouhet	$\sum_{n < 2^m} \varepsilon_n p(n) = 0$ when $\deg p < m.$
Sharp moment	$\sum_{n < 2^m} \varepsilon_n n^m = (-1)^m m! 2^{\binom{m}{2}}.$
Sparse Prouhet	$\sum_{k \preceq 2n} \varepsilon_k p(k) = 0$ when $\deg p < \text{wt}_2(n).$
Prefix discrepancy	$\sum_{n < N} \varepsilon_n = 0$ for even N , $\varepsilon_{(N-1)/2}$ for odd $N.$
Walsh transform	A length- 2^m block is one Walsh frequency.
Ordinary DFT	$\widehat{\varepsilon}_m(k) = \prod_{j < m} (1 - e^{-2\pi i k 2^j / 2^m}).$
Riesz density	$2^{-m} P_m(e^{2\pi i x}) ^2 = \prod_{j < m} (1 - \cos(2\pi 2^j x)).$
Characteristic two	$(1+x)^3 T(x)^2 + (1+x)^2 T(x) + x = 0$ in $\mathbb{F}_2[[x]].$
Weighted enumerator	$\sum_n u^{\text{wt}_2(n)} z^n = \prod_j (1 + u z^{2^j}).$
Subtraction cocycle	$\varepsilon_{a-b} = \varepsilon_a \varepsilon_b (-1)^{\nu_2 \binom{a}{b}}$ for $b \leq a.$
Autocorrelation Mahler	$2zC(z) = 1 - (1-z)^2 C(z^2).$
Natural boundary	E continues across no arc of $ z = 1.$
Hankel rigidity	$\det(\varepsilon_{i+j})_{i,j < r} \neq 0$ for every $r.$
Rarefaction mod 3	$(A_{2u,r}^{(3)})_r = 3^{u-1} (2, -1, -1).$
Gelfond exponent	$3^{m/2} \leq \sup_x P_m(e^{2\pi i x}) \leq \frac{2}{\sqrt{3}} 3^{m/2}.$
Newman positivity	$\sum_{n < X, 3 n} \varepsilon_n > 0$ for all $X \geq 1.$
Critical exponent	Overlap-free; squares occur, exponent exactly 2.
Factor complexity	$\ell + 1 \leq p(\ell) < 8\ell.$
Dirichlet–Mellin	$D(s, a) = \Gamma(s)^{-1} \int_0^\infty t^{s-1} e^{-at} E(e^{-t}) dt.$

2 Binary character, blocks, automata, and enumerators

2.1 Recursion and uniqueness

Theorem 2.1 (Binary recursion and uniqueness). *The signed sequence satisfies*

$$\varepsilon_0 = 1, \quad \varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n. \quad (2.1)$$

Equivalently,

$$\tau(0) = 0, \quad \tau(2n) = \tau(n), \quad \tau(2n+1) = 1 - \tau(n). \quad (2.2)$$

Either system and its initial value determines the sequence uniquely.

Proof. Appending a terminal binary digit 0 leaves the digit sum unchanged; appending a terminal 1 increases it by one. For uniqueness, every positive integer is $2q$ or $2q+1$ with $q < n$, so strong induction eventually reduces its value to the initial value at zero. $\square$

Let $W_m = \tau(0)\tau(1) \cdots \tau(2^m - 1)$, and let $\overline{W}$ denote bitwise complement. The recurrence gives

$$W_{m+1} = W_m \overline{W_m}. \quad (2.3)$$

Thus the infinite word is the fixed point beginning with 0 of

$$0 \mapsto 01, \quad 1 \mapsto 10. \quad (2.4)$$

In signed form a block is followed by its negative.

2.2 Binary concatenation and reflection

Theorem 2.2 (Block concatenation). *For $m, q \geq 0$ and $0 \leq r < 2^m$,*

$$\text{wt}_2(2^m q + r) = \text{wt}_2(q) + \text{wt}_2(r), \quad (2.5)$$

$$\varepsilon_{2^m q + r} = \varepsilon_q \varepsilon_r, \quad (2.6)$$

$$\tau(2^m q + r) = \tau(q) \oplus \tau(r). \quad (2.7)$$

Proof. Multiplication by 2^m shifts the binary expansion of q by m places. Since $r < 2^m$, adding r fills only those vacated positions and creates no carry. The bits concatenate, so weights add and parities follow. $\square$

Corollary 2.3 (Reflection and bit-position symmetry). *For $0 \leq n < 2^m$,*

$$\varepsilon_{2^m - 1 - n} = (-1)^m \varepsilon_n, \quad \tau(2^m - 1 - n) = \tau(n) \oplus (m \bmod 2). \quad (2.8)$$

Permuting the first m binary positions leaves ε_n and $\tau(n)$ unchanged.

Proof. Within m positions, $2^m - 1 - n$ is the bitwise complement of n , so its weight is $m - \text{wt}_2(n)$. A permutation of positions leaves their sum unchanged. $\square$

2.3 Exclusive-or is the natural group law

Theorem 2.4 (XOR character). *For all $a, b \geq 0$,*

$$\tau(a \oplus b) = \tau(a) \oplus \tau(b), \quad \boxed{\varepsilon_{a \oplus b} = \varepsilon_a \varepsilon_b}. \quad (2.9)$$

Thus $n \mapsto \varepsilon_n$ is a character of the Boolean group $\bigoplus_{j \geq 0} \mathbb{Z}/2\mathbb{Z}$.

Proof. At each position, XOR adds the two bits modulo two. Summing the coordinate identities modulo two gives $\text{wt}_2(a \oplus b) \equiv \text{wt}_2(a) + \text{wt}_2(b) \pmod{2}$. $\square$

Corollary 2.5 (XOR autocorrelation and Hamming spheres). *Fix $m \geq 0$, $0 \leq a < 2^m$, and $0 \leq r \leq m$. Then*

$$\sum_{n=0}^{2^m-1} \varepsilon_n \varepsilon_{n \oplus a} = 2^m \varepsilon_a, \quad (2.10)$$

$$\sum_{\substack{0 \leq n < 2^m \\ \text{wt}_2(n \oplus a) = r}} \varepsilon_n = \varepsilon_a (-1)^r \binom{m}{r}. \quad (2.11)$$

Proof. The first summand is termwise ε_a . For the second, put $h = n \oplus a$. Then $\varepsilon_n = \varepsilon_a \varepsilon_h = \varepsilon_a (-1)^r$, and there are $\binom{m}{r}$ masks of weight r . $\square$

Formal versions. `Fabius.sum_thueMorseSign_mul_xor` (the autocorrelation, which the Lean proof establishes termwise from the character identity, with no alignment hypothesis on a) and `Fabius.sum_thueMorseSign_hammingSphere` (the sphere sums, by reindexing along the XOR involution), in `ThueMorseEnumerators.lean`.

2.4 Finite automata, matrices, and tensor powers

The recurrence gives a two-state automaton: begin in state $+1$, read the binary digits in either order, and flip the state for each one-bit. In matrix form, let

$$M_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad M_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad e_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (2.12)$$

If $n = \sum_{j=0}^{m-1} b_j 2^j$, then

$$M_{b_0} \cdots M_{b_{m-1}} e_0 = \begin{pmatrix} 1 - \tau(n) \\ \tau(n) \end{pmatrix}. \quad (2.13)$$

The matrices commute, reflecting the fact that only the number of one-bits matters. In natural binary order,

$$(\varepsilon_0, \dots, \varepsilon_{2^m-1})^\top = ((1, -1)^\top)^{\otimes m}. \quad (2.14)$$

Hence every residue subsequence in the 2-kernel is either ε or $-\varepsilon$:

$$(\varepsilon_{2^m n+r})_{n \geq 0} = \varepsilon_r (\varepsilon_n)_{n \geq 0}. \quad (2.15)$$

This is the finite-kernel criterion for 2-automaticity in its smallest possible form.

Algorithm 2.6 (XOR-fold evaluator). For a fixed-width machine word, fold the bits by XORing with successively larger right shifts. The final least significant bit is $\tau(n)$.

```
x := n
x := x xor (x >> 1)
x := x xor (x >> 2)
x := x xor (x >> 4)
x := x xor (x >> 8)
... continue while the shift is below the word width ...
tau := x & 1
epsilon := 1 - 2*tau
```

2.5 Exact weight distribution in a dyadic block

The four drafts repeatedly use the fact that a length- 2^m block is a Boolean cube. It is useful to record the complete distribution, not only its parity consequence.

Theorem 2.7 (Binomial weight enumerator). *For every $m \geq 0$,*

$$\sum_{n=0}^{2^m-1} u^{\text{wt}_2(n)} = (1+u)^m. \quad (2.16)$$

Consequently

$$\#\{0 \leq n < 2^m : \text{wt}_2(n) = r\} = \binom{m}{r}. \quad (2.17)$$

Under the uniform distribution on the block, $\text{wt}_2(n)$ is binomial with mean $m/2$ and variance $m/4$.

Proof. Each of the m positions independently contributes 1 or u . Multiplication gives $(1+u)^m$; coefficient extraction and the standard binomial moments give the remaining statements. $\square$

Formal versions. The enumerator is `Fabius.sum_pow_binaryWeight_eq_one_add_pow`, proved over an arbitrary commutative semiring, and the count (2.17) is `Fabius.card_filter_binaryWeight_eq`, extracted from it exactly as in the proof above: read the enumerator in $\mathbb{N}[z]$ and compare the coefficient of z^r . Both live in `ThueMorseEnumerators.lean`; the underlying Boolean-cube machinery is described after [theorem 6.2](#).

Setting $u = -1$ gives exact sign balance for $m \geq 1$. More generally, the characteristic function of the weight is

$$2^{-m} \sum_{n < 2^m} e^{it \text{wt}_2(n)} = \left(\frac{1 + e^{it}}{2} \right)^m. \quad (2.18)$$

This elementary statistical layer is useful when comparing Boolean/Walsh structure with ordinary Fourier structure later in the report.

2.6 Evil and odious enumerators

Call n *evil* when $\text{wt}_2(n)$ is even and *odious* when it is odd. Let $\mathcal{E}_n$ and $\mathcal{O}_n$ be the increasing enumerations, starting at index zero.

Theorem 2.8 (Evil and odious numbers). *For every $n \geq 0$,*

$$\boxed{\mathcal{E}_n = 2n + \tau(n), \quad \mathcal{O}_n = 2n + 1 - \tau(n).} \quad (2.19)$$

Hence

$$\mathcal{O}_n - \mathcal{E}_n = \varepsilon_n, \quad \mathcal{O}_n + \mathcal{E}_n = 4n + 1. \quad (2.20)$$

Proof. The two consecutive integers $2n$ and $2n + 1$ have parities $\tau(n)$ and $1 - \tau(n)$ of their binary weights. If $\tau(n) = 0$, then $2n$ is evil and $2n + 1$ odious; if $\tau(n) = 1$, the roles reverse. The displayed formulas choose the correct member of each consecutive pair. $\square$

Corollary 2.9 (Composition laws). *For every $n \geq 0$,*

$$\mathcal{E}_{\mathcal{E}_n} = 2\mathcal{E}_n, \quad \mathcal{O}_{\mathcal{E}_n} = 2\mathcal{E}_n + 1, \quad (2.21)$$

$$\mathcal{E}_{\mathcal{O}_n} = 2\mathcal{O}_n + 1, \quad \mathcal{O}_{\mathcal{O}_n} = 2\mathcal{O}_n. \quad (2.22)$$

Proof. By definition $\tau(\mathcal{E}_n) = 0$ and $\tau(\mathcal{O}_n) = 1$. Substitute these values into (2.19) with the new index. $\square$

Formal versions. In `ThueMorseEnumerators.lean` the enumerations are `Fabius.evileNum` and `Fabius.odiousEnum`, with landing lemmas `Fabius.thueMorseBit_evilEnum`, `Fabius.thueMorseBit_odiousEnum`, strict monotonicity `Fabius.evileNum_strictMono`, `Fabius.odiousEnum_strictMono`, difference and sum `Fabius.odiousEnum_sub_evilEnum`, `Fabius.evileNum_add_odiousEnum`, and the four composition laws. The Lean proof of surjectivity isolates a detail worth recording in prose: for every m with $\tau(m) = 0$ one has $\mathcal{E}_{\lfloor m/2 \rfloor} = m$ (`Fabius.evileNum_div_two`) – the same index $\lfloor m/2 \rfloor$ works whether m is even (then $\tau(m/2) = 0$ and the τ -summand vanishes) or odd (then $\tau((m-1)/2) = 1$ and the summand supplies the missing one) – and dually for odious m (`Fabius.odiousEnum_div_two`). Monotone enumerations that hit everything are complete, so no separate counting argument is needed.

3 Digit, floor, fractional-part, and trigonometric formulas

3.1 Extracting one binary digit

Lemma 3.1 (Floor extraction). *For every $j, n \geq 0$,*

$$b_j(n) = \left\lfloor \frac{n}{2^j} \right\rfloor - 2 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor. \quad (3.1)$$

Proof. Write $\lfloor n/2^j \rfloor = 2q + r$ with $r \in \{0, 1\}$. Then $q = \lfloor n/2^{j+1} \rfloor$, while r is exactly the j -th binary digit. $\square$

Since $(-1)^b = 1 - 2b$ for $b \in \{0, 1\}$, the definition factorizes over coordinates.

Theorem 3.2 (Digit product and Boolean polynomial). *For every $n \geq 0$,*

$$\varepsilon_n = \prod_{j \geq 0} (1 - 2b_j(n)), \quad \tau(n) = \frac{1 - \prod_{j \geq 0} (1 - 2b_j(n))}{2}. \quad (3.2)$$

Only finitely many factors differ from 1. If J is any finite set containing the support of the binary expansion, then

$$\tau(n) = \sum_{\emptyset \neq S \subseteq J} (-2)^{|S|-1} \prod_{j \in S} b_j(n). \quad (3.3)$$

Proof. Multiply $(-1)^{b_j} = 1 - 2b_j$ over all positions. Expanding the product in the zero–one conversion (1.3) gives (3.3). $\square$

Substitution of (3.1) gives a digit-free, floor-only product:

$$\varepsilon_n = \prod_{j \geq 0} \left(1 - 2 \left\lfloor \frac{n}{2^j} \right\rfloor + 4 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor \right). \quad (3.4)$$

Every factor is $1 - 2b_j(n) \in \{1, -1\}$.

3.2 Dyadic floor and fractional-part sums

Lemma 3.3 (Legendre floor sum). *For every $n \geq 0$,*

$$\sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = n - \text{wt}_2(n), \quad (3.5)$$

$$\sum_{j \geq 0} \left\lfloor \frac{n}{2^j} \right\rfloor = 2n - \text{wt}_2(n). \quad (3.6)$$

Proof. Insert the binary expansion and interchange finite sums:

$$\begin{aligned} \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor &= \sum_{j \geq 1} \sum_{\ell \geq j} b_\ell(n) 2^{\ell-j} \\ &= \sum_{\ell \geq 1} b_\ell(n) (2^\ell - 1) = n - \sum_{\ell \geq 0} b_\ell(n). \end{aligned}$$

Adding the $j = 0$ term gives the second identity. $\square$

Theorem 3.4 (Floor-sum formulae). *For every $n \geq 0$,*

$$\varepsilon_n = (-1)^{\sum_{j \geq 0} \lfloor n/2^j \rfloor}, \quad \tau(n) \equiv \sum_{j \geq 0} \left\lfloor \frac{n}{2^j} \right\rfloor \pmod{2}. \quad (3.7)$$

Equivalently,

$$\varepsilon_n = \prod_{j \geq 0} (-1)^{\lfloor n/2^j \rfloor}. \quad (3.8)$$

Proof. By (3.6), the exponent differs from $\text{wt}_2(n)$ by an even integer. $\square$

Formal versions. `ThueMorseDigits.lean` proves the additive Legendre formula $\text{wt}_2(n) + \nu_2(n!) = n$ (`Fabius.binaryWeight_add_padicValNat_factorial`), the floor sums in truncation-free form (`Fabius.binaryWeight_add_sum_div_two_pow`, `Fabius.sum_div_two_pow_add_binaryWeight`), the floor-parity formula (3.7) (`Fabius.thueMorseSign_eq_neg_one_pow_sum_div`), the bound $\text{wt}_2(n) \leq n$ (`Fabius.binaryWeight_le_self`), and the bit-extraction bridge $\text{wt}_2(n) = \sum_{j < m} b_j(n)$ (`Fabius.binaryWeight_eq_sum_testBit`) together with the digit product (3.2) (`Fabius.thueMorseSign_eq_prod_testBit`). Natural-number subtraction being awkward, every Lean statement is phrased additively; the prose forms above are recovered by moving one term across.

There is also a genuinely infinite real-valued representation whose sum is nevertheless always an integer.

Proposition 3.5 (Fractional-part digit sum). *For every $n \geq 0$,*

$$\boxed{\text{wt}_2(n) = \sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\}}. \quad (3.9)$$

Consequently

$$\varepsilon_n = \cos \left(\pi \sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\} \right), \quad \tau(n) = \frac{1 - \varepsilon_n}{2}. \quad (3.10)$$

Proof. The geometric series gives $\sum_{j \geq 1} n/2^j = n$. Subtract (3.5) termwise:

$$\sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\} = n - \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = \text{wt}_2(n).$$

□

3.3 Square waves and a sine-sign product

For nonintegral x ,

$$\text{sgn}(\sin \pi x) = (-1)^{\lfloor x \rfloor}. \quad (3.11)$$

The half-unit shift avoids every zero.

Theorem 3.6 (Rademacher–sine formula). *For every $n \geq 0$,*

$$\boxed{\varepsilon_n = \prod_{j \geq 0} \text{sgn} \left(\sin \frac{\pi(n + \frac{1}{2})}{2^j} \right)}. \quad (3.12)$$

All sufficiently large factors are +1. Moreover the j -th factor is exactly $(-1)^{b_j(n)}$.

Proof. The number $(n + \frac{1}{2})/2^j$ is never integral and has the same floor as $n/2^j$. Apply (3.11), then observe from (3.1) that $(-1)^{\lfloor n/2^j \rfloor} = (-1)^{b_j(n)}$. The factors are positive once $2^j > n + \frac{1}{2}$. □

3.4 The sign of the Fabius sinc product

Define $\text{sinc } y = \sin y/y$, with $\text{sinc } 0 = 1$, and

$$\Phi(x) = \prod_{j=0}^{\infty} \text{sinc} \left(\frac{\pi x}{2^j} \right). \quad (3.13)$$

The product converges locally uniformly because the logarithm of the tail is controlled by $\sum_j x^2/4^j$. It is, up to normalization conventions, the infinite Fourier product naturally associated with the Rvachev up-function.

Corollary 3.7 (Interval sign of the sinc product). *If $N \in \mathbb{N}$ and $N < x < N + 1$, then*

$$\operatorname{sgn} \Phi(x) = \varepsilon_N. \quad (3.14)$$

In particular,

$$\varepsilon_n = \operatorname{sgn} \Phi\left(n + \frac{1}{2}\right). \quad (3.15)$$

Proof. For $x > 0$, every sinc denominator is positive. No multiple of 2^j lies strictly between the consecutive integers N and $N + 1$, so $\lfloor x/2^j \rfloor = \lfloor N/2^j \rfloor$. The numerator signs therefore multiply to $(-1)^{\sum_j \lfloor N/2^j \rfloor} = \varepsilon_N$ by [theorem 3.4](#). $\square$

3.5 The ruler transition

Incrementing n flips all trailing one-bits to zero and the next zero-bit to one. The number of erased one-bits is $\nu_2(n + 1)$.

Theorem 3.8 (Successor and ruler formulae). *For every $n \geq 0$,*

$$\operatorname{wt}_2(n + 1) - \operatorname{wt}_2(n) = 1 - \nu_2(n + 1), \quad (3.16)$$

$$\varepsilon_{n+1} = (-1)^{1+\nu_2(n+1)} \varepsilon_n, \quad (3.17)$$

$$(-1)^{\nu_2(n+1)} = -\varepsilon_n \varepsilon_{n+1}. \quad (3.18)$$

Consequently

$$\varepsilon_n = \prod_{k=1}^n (-1)^{1+\nu_2(k)}. \quad (3.19)$$

Proof. The bit transition changes the weight by one minus the number of trailing ones. Exponentiate (-1) , then telescope from $\varepsilon_0 = 1$. $\square$

Formal versions. The successor law in additive form $\operatorname{wt}_2(n + 1) + \nu_2(n + 1) = \operatorname{wt}_2(n) + 1$ is `Fabius.binaryWeight_succ_add_padicValNat`; the adjacent-sign readout (3.18) is `Fabius.thueMorseSign_mul_succ`, the recursion (3.17) is `Fabius.thueMorseSign_succ_eq`, and the telescoped ruler product (3.19) is `Fabius.thueMorseSign_eq_prod_ruler`, all in `ThueMorseValuation.lean`.

4 Factorial valuations, central binomials, Catalan numbers, and carries

4.1 Legendre’s formula

The floor sum in [theorem 3.3](#) counts powers of two in a factorial.

Theorem 4.1 (Factorial valuation). *For every $n \geq 0$,*

$$\nu_2(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = n - \operatorname{wt}_2(n). \quad (4.1)$$

Hence

$$\varepsilon_n = (-1)^{n-\nu_2(n!)} = (-1)^{n+\nu_2(n!)}, \quad \tau(n) \equiv n + \nu_2(n!) \pmod{2}. \quad (4.2)$$

Proof. Every multiple of 2 contributes at least one factor, every multiple of 4 contributes one additional factor, and so on. This gives the floor sum; [theorem 3.3](#) gives the digit form. Addition and subtraction agree modulo two. $\square$

The telescoping product (3.19) returns to this formula because $\sum_{k=1}^n \nu_2(k) = \nu_2(n!)$.

4.2 A single central binomial coefficient

Theorem 4.2 (Central-binomial valuation). *For every $n \geq 0$,*

$$\boxed{\nu_2 \binom{2n}{n} = \text{wt}_2(n).} \quad (4.3)$$

Consequently

$$\boxed{\varepsilon_n = (-1)^{\nu_2 \binom{2n}{n}}, \quad \tau(n) = \nu_2 \binom{2n}{n} \bmod 2.} \quad (4.4)$$

Equivalently,

$$\tau(n) = \sin^2 \left(\frac{\pi}{2} \nu_2 \binom{2n}{n} \right), \quad \varepsilon_n = \cos \left(\pi \nu_2 \binom{2n}{n} \right). \quad (4.5)$$

Proof. Use (4.1) and $\text{wt}_2(2n) = \text{wt}_2(n)$:

$$\begin{aligned} \nu_2 \binom{2n}{n} &= \nu_2((2n)!) - 2\nu_2(n!) \\ &= (2n - \text{wt}_2(2n)) - 2(n - \text{wt}_2(n)) = \text{wt}_2(n). \end{aligned}$$

□

The shortest Pascal replacement. The source formula inspects all $n + 1$ entries of row n and takes an exact logarithm. Formula (4.4) inspects the two-adic order of one entry in row $2n$, and in fact recovers the entire digit sum rather than only its parity.

4.3 Catalan valuations

Let

$$\text{Cat}_n = \frac{1}{n+1} \binom{2n}{n}. \quad (4.6)$$

Corollary 4.3 (Catalan formula). *For every $n \geq 0$,*

$$\nu_2(\text{Cat}_n) = \text{wt}_2(n+1) - 1, \quad (4.7)$$

and hence

$$\boxed{\varepsilon_{n+1} = -(-1)^{\nu_2(\text{Cat}_n)}.} \quad (4.8)$$

Proof. By theorem 4.2, $\nu_2(\text{Cat}_n) = \text{wt}_2(n) - \nu_2(n+1)$. Equation (3.16) rewrites this as $\text{wt}_2(n+1) - 1$. □

Formal versions. `Fabius.catalan_padicValNat_two_add_one` (stated additively as $\nu_2(\text{Cat}_n) + 1 = \text{wt}_2(n+1)$) and the sign form `Fabius.thueMorseSign_succ_eq_neg_catalan`, in `ThueMorseValuation.lean`. The Lean proof derives the positivity of Catalan numbers inline from $(n+1) \text{Cat}_n = \binom{2n}{n}$ rather than importing it.

4.4 Kummer’s carry cocycle

Theorem 4.4 (Addition with carries). *For all $a, b \geq 0$,*

$$\nu_2 \binom{a+b}{a} = \text{wt}_2(a) + \text{wt}_2(b) - \text{wt}_2(a+b). \quad (4.9)$$

This is the number of carry operations in the binary addition of a and b , including propagated carries. Therefore

$$\boxed{\varepsilon_{a+b} = \varepsilon_a \varepsilon_b (-1)^{\nu_2 \binom{a+b}{a}}.} \quad (4.10)$$

Equivalently,

$$\tau(a+b) \equiv \tau(a) + \tau(b) + \nu_2 \binom{a+b}{a} \pmod{2}. \quad (4.11)$$

Proof. Legendre’s formula gives

$$\nu_2 \binom{a+b}{a} = (a+b - \text{wt}_2(a+b)) - (a - \text{wt}_2(a)) - (b - \text{wt}_2(b)),$$

which is (4.9). Rearranging modulo two and raising (-1) to both sides gives the sign formula. The carry interpretation is the binary case of Kummer’s theorem. $\square$

The no-carry case $a \& b = 0$ makes the valuation zero and recovers ordinary multiplicativity. In particular, [theorem 2.2](#) is the separated-support case.

Corollary 4.5 (Multinomial carry formula). *For $a_1, \dots, a_r \geq 0$ and $N = a_1 + \dots + a_r$,*

$$\nu_2 \binom{N}{a_1, \dots, a_r} = \sum_{j=1}^r \text{wt}_2(a_j) - \text{wt}_2(N), \quad (4.12)$$

$$\varepsilon_N = \left(\prod_{j=1}^r \varepsilon_{a_j} \right) (-1)^{\nu_2 \binom{N}{a_1, \dots, a_r}}. \quad (4.13)$$

Proof. Apply Legendre’s formula to $N!/(a_1! \cdots a_r!)$ and then take parity. $\square$

4.5 Subtraction and borrow counts

Kummer’s theorem has a mirror statement for subtraction. For $0 \leq b \leq a$, the same Legendre computation applied to $\binom{a}{b} = a!/(b!(a-b)!)$ gives

$$\nu_2 \binom{a}{b} = \text{wt}_2(b) + \text{wt}_2(a-b) - \text{wt}_2(a), \quad (4.14)$$

which counts the borrow operations, including propagated borrows, in the binary subtraction $a - b$. Therefore

$$\boxed{\varepsilon_{a-b} = \varepsilon_a \varepsilon_b (-1)^{\nu_2 \binom{a}{b}}.} \quad (4.15)$$

Addition and subtraction thus carry the same sign correction, read from the two-adic valuation of the binomial coefficient whose top entry is the larger of the numbers involved; (4.10) is recovered from (4.15) by renaming $a + b$ as the top entry.

Formal versions. Kummer’s carry formula and cocycle were already in the corpus (`Fabius.addChoose_padicValNat_two`, `Fabius.thueMorseSign_add_valuation`, in `TwoAdic.lean`); `ThueMorseValuation.lean` adds the truncation-free additive form `Fabius.binaryWeight_add_addChoose_padicValNat`, the weight subadditivity $\text{wt}_2(a+b) \leq \text{wt}_2(a) + \text{wt}_2(b)$ (`Fabius.binaryWeight_add_le`, itself proved by a four-way parity induction), and the borrow pair `Fabius.subChoose_padicValNat_two_add`, `Fabius.thueMorseSign_sub_valuation`.

5 Pascal support, residue formulae, multinomials, and Boolean inversion

5.1 Lucas's theorem as a submask criterion

Theorem 5.1 (Lucas parity criterion). *For $0 \leq k \leq n$,*

$$\binom{n}{k} \equiv 1 \pmod{2} \iff k \preccurlyeq_2 n. \quad (5.1)$$

Equivalently,

$$\mathbf{1}_{k \preccurlyeq_2 n} = \frac{1 - (-1)^{\binom{n}{k}}}{2}. \quad (5.2)$$

Proof. In $\mathbb{F}_2[z]$, Frobenius gives

$$(1+z)^n = \prod_{j: b_j(n)=1} (1+z)^{2^j} = \prod_{j: b_j(n)=1} (1+z^{2^j}).$$

The exponents occurring in the expansion are exactly the submasks of n , and their coefficients are one modulo two. $\square$

Define the odd-support polynomial of row n by

$$Q_n(z) = \sum_{k=0}^n \mathbf{1}_{\binom{n}{k} \text{ odd}} z^k. \quad (5.3)$$

The proof gives the complete factorization

$$Q_n(z) = \prod_{j: b_j(n)=1} (1 + z^{2^j}). \quad (5.4)$$

Corollary 5.2 (Glaisher–Fine count). *The number $\nu(n)$ of odd entries in row n is*

$$\nu(n) = 2^{\text{wt}_2(n)}. \quad (5.5)$$

Proof. Set $z = 1$ in (5.4). $\square$

The source formula is now transparent. Since even entries contribute $+1$ and odd entries contribute -1 ,

$$\mathcal{X}(n) = n + 1 - 2\nu(n) = n + 1 - 2^{\text{wt}_2(n)+1}, \quad (5.6)$$

which is exactly (1.5).

5.2 A logarithm-free Pascal-row formula modulo three

Let $\text{rem}_3(x) \in \{0, 1, 2\}$ denote the least nonnegative residue modulo three. Powers of two alternate between one and two modulo three.

Theorem 5.3 (Odd-count residue formula). *For every $n \geq 0$,*

$$\tau(n) = \text{rem}_3(\nu(n)) - 1, \quad \varepsilon_n = 3 - 2\text{rem}_3(\nu(n)). \quad (5.7)$$

The residue is always 1 or 2, never 0.

Proof. By (5.5), $\nu(n) = 2^{\text{wt}_2(n)}$. Its residue is one when $\text{wt}_2(n)$ is even and two when it is odd. $\square$

A form using exactly the row signs of the source formula can be written without the odd-count notation. Put

$$S(n) = \sum_{k=1}^n (-1)^{\binom{n}{k}}, \quad S(0) = 0. \quad (5.8)$$

Since the omitted $k = 0$ term is -1 ,

$$S(n) = n + 2 - 2^{\text{wt}_2(n)+1}. \quad (5.9)$$

Corollary 5.4 (Binomial residue and sine formulae). *For every $n \geq 0$,*

$$\tau(n) = \text{rem}_3(2n + S(n)), \quad (5.10)$$

$$\tau(n) = \frac{4}{3} \sin^2\left(\frac{\pi}{3}(n - S(n))\right). \quad (5.11)$$

Proof. Equation (5.9) gives $2n + S(n) = 3n + 2 - 2^{\text{wt}_2(n)+1}$, whose residue is zero or one according as the weight is even or odd. Also $n - S(n) = 2^{\text{wt}_2(n)+1} - 2$, congruent to zero or two modulo three; the squared sine is therefore zero or $3/4$. $\square$

The logarithmic and residue formulas are two shadows of the same invariant $\nu(n)$: the first reads the exponent of a power of two; the second reads that exponent's parity from the two-cycle of powers modulo three.

5.3 A one-parameter row statistic

For a scalar u , define

$$R_u(n) = \sum_{k=0}^n u^{\binom{n}{k} \bmod 2}. \quad (5.12)$$

Exactly $\nu(n)$ exponents are one and the others are zero, so

$$R_u(n) = n + 1 + (u - 1)2^{\text{wt}_2(n)}. \quad (5.13)$$

Thus $R_{-1}(n) = \mathcal{X}(n)$, while $R_0(n)$ counts the even entries. For every fixed $u \neq 1$,

$$2^{\text{wt}_2(n)} = \frac{R_u(n) - (n + 1)}{u - 1}. \quad (5.14)$$

The signed choice $u = -1$ is elegant, but structurally it is one member of a whole exact family.

5.4 Sparse Pascal probes for individual bits

Lucas's criterion at $k = 2^j$ isolates one binary coordinate:

$$\binom{n}{2^j} \equiv b_j(n) \pmod{2}. \quad (5.15)$$

Theorem 5.5 (Power-of-two column formula). *For every $n \geq 0$,*

$$\varepsilon_n = (-1)^{\sum_{j \geq 0} \binom{n}{2^j}} = \prod_{j \geq 0} \cos\left(\pi \binom{n}{2^j}\right), \quad (5.16)$$

$$\tau(n) = \frac{1}{2} \left(1 - \prod_{j \geq 0} \cos\left(\pi \binom{n}{2^j}\right) \right). \quad (5.17)$$

Only the columns $2^j \leq n$ matter.

Proof. Only the parity of each binomial coefficient contributes to the exponent or cosine. Equation (5.15) turns their parity sum into $\text{wt}_2(n)$. $\square$

Unlike the source formula, this scans only $O(\log n)$ selected entries and uses no logarithm. Its price is the distinguished set of power-of-two columns.

5.5 Odd multinomial coefficients and exact r -nomial logarithms

Fix $r \geq 2$. Let $O_r(n)$ be the number of odd coefficients in $(x_1 + \cdots + x_r)^n$.

Theorem 5.6 (Multinomial support count). *For every $n \geq 0$,*

$$\boxed{O_r(n) = r^{\text{wt}_2(n)}}. \quad (5.18)$$

Consequently

$$\boxed{\varepsilon_n = (-1)^{\log_r O_r(n)}, \quad \tau(n) = \frac{1 - (-1)^{\log_r O_r(n)}}{2}}. \quad (5.19)$$

The logarithm is exact.

Proof. In characteristic two,

$$(x_1 + \cdots + x_r)^n = \prod_{j: b_j(n)=1} (x_1^{2^j} + \cdots + x_r^{2^j}). \quad (5.20)$$

For each one-bit of n , choose independently one of the r variables to receive it. Unique binary expansions make distinct assignments produce distinct exponent vectors, so there are $r^{\text{wt}_2(n)}$ nonzero coefficients modulo two. $\square$

There are $\binom{n+r-1}{r-1}$ weak compositions of n into r parts. Define

$$\mathcal{X}_r(n) = \sum_{k_1 + \cdots + k_r = n} (-1)^{\binom{n}{k_1, \dots, k_r}}. \quad (5.21)$$

Corollary 5.7 (Signed r -nomial row formula). *For every $r \geq 2$,*

$$\binom{n+r-1}{r-1} - \mathcal{X}_r(n) = 2r^{\text{wt}_2(n)}, \quad (5.22)$$

and therefore

$$\boxed{\tau(n) = \frac{1 - (-1)^{\log_r \left(\frac{\binom{n+r-1}{r-1} - \mathcal{X}_r(n)}{2} \right)}}{2}}. \quad (5.23)$$

Proof. Each odd multinomial coefficient contributes -1 , each even coefficient $+1$. Subtracting the signed sum from the total count gives twice the odd support size, which is $2r^{\text{wt}_2(n)}$ by theorem 5.6. $\square$

5.6 Boolean weight enumerators and Möbius inversion

The submasks of n form a Boolean lattice on its set of one-bit positions.

Theorem 5.8 (Submask weight enumerator). *For every indeterminate z ,*

$$\sum_{k \preceq_2 n} z^{\text{wt}_2(k)} = (1 + z)^{\text{wt}_2(n)}, \quad (5.24)$$

$$z^{\text{wt}_2(n)} = \sum_{k \preceq_2 n} (z - 1)^{\text{wt}_2(k)}. \quad (5.25)$$

More generally,

$$\sum_{k \preceq_2 n} x^{\text{wt}_2(k)} y^{\text{wt}_2(n-k)} = (x + y)^{\text{wt}_2(n)}. \quad (5.26)$$

Proof. For each occupied bit of n , choose whether it belongs to k , to its complement $n - k$, or is omitted in the one-variable specialization. The second formula is the first with z replaced by $z - 1$. $\square$

Important specializations are

$$\#\{k : k \preceq_2 n\} = 2^{\text{wt}_2(n)}, \quad (5.27)$$

$$\sum_{k \preceq_2 n} \varepsilon_k = \begin{cases} 1, & n = 0, \\ 0, & n > 0, \end{cases} \quad (5.28)$$

$$\varepsilon_n = \sum_{k \preceq_2 n} (-2)^{\text{wt}_2(k)}. \quad (5.29)$$

Theorem 5.9 (Boolean Möbius inversion). *If $k \preceq_2 n$, then the Möbius function of the submask poset is*

$$\mu_2(k, n) = (-1)^{\text{wt}_2(n) - \text{wt}_2(k)} = \varepsilon_{n \oplus k}. \quad (5.30)$$

Thus

$$g(n) = \sum_{k \preceq_2 n} f(k) \iff f(n) = \sum_{k \preceq_2 n} \varepsilon_{n \oplus k} g(k). \quad (5.31)$$

Proof. The interval $[k, n]$ is the Boolean lattice on the bits of n absent from k ; its rank is $\text{wt}_2(n) - \text{wt}_2(k) = \text{wt}_2(n \oplus k)$. The usual Boolean Möbius function is $(-1)^{\text{rank}}$, and standard Möbius inversion applies. $\square$

Let

$$P_{n,k} = \mathbf{1}_{\binom{n}{k} \text{ odd}} = \mathbf{1}_{k \preceq_2 n}. \quad (5.32)$$

As a locally finite lower-triangular matrix,

$$(P^{-1})_{n,k} = P_{n,k} \varepsilon_{n \oplus k}. \quad (5.33)$$

In particular, the first column of P^{-1} is the signed Thue–Morse sequence. This is a second global Pascal encoding: ε is not only a row statistic but the Möbius inverse of the entire parity incidence matrix.

Combining (5.29), Lucas’s indicator, and the central binomial valuation gives a formula involving only ordinary binomial coefficients:

Theorem 5.10 (Nested binomial formula). *For every $n \geq 0$,*

$$\varepsilon_n = \sum_{k=0}^n \frac{1 - (-1)^{\binom{n}{k}}}{2} (-2)^{\nu_2 \binom{2k}{k}}. \quad (5.34)$$

Proof. The first factor is $\mathbf{1}_{k \preceq_2 n}$ by (5.2), and the exponent in the second is $\text{wt}_2(k)$ by theorem 4.2. The sum is therefore exactly (5.29). $\square$

6 Finite and infinite generating products

6.1 The Boolean-cube mother identity

A multivariate identity contains the finite generating product, the weight enumerator, and the Walsh character formula as specializations.

Theorem 6.1 (Boolean-cube product). *For commuting indeterminates $x_0, \dots, x_{m-1}$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n \prod_{j=0}^{m-1} x_j^{b_j(n)} = \prod_{j=0}^{m-1} (1 - x_j). \quad (6.1)$$

Proof. Identify n with its bit vector $b \in \{0, 1\}^m$. The sum factors over coordinates:

$$\sum_{b \in \{0,1\}^m} \prod_j (-x_j)^{b_j} = \prod_j (1 - x_j).$$

□

Setting all $x_j = -u$ recovers (2.16); setting $x_j = z^{2^j}$ gives the central finite polynomial.

6.2 The finite Prouhet polynomial

Define

$$P_m(z) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n. \quad (6.2)$$

Theorem 6.2 (Complete finite factorization). *For every $m \geq 0$,*

$$P_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}). \quad (6.3)$$

The empty product for $m = 0$ is 1.

Proof. Use theorem 6.1 with $x_j = z^{2^j}$. Equivalently, expanding the product chooses a subset of powers of two; its sum is an integer $n < 2^m$, uniquely, and the coefficient is $(-1)^{\#\text{chosen bits}} = \varepsilon_n$. □

Corollary 6.3 (Coefficient and block formulae). *For every $n \geq 0$,*

$$\varepsilon_n = [z^n] \prod_{j \geq 0} (1 - z^{2^j}), \quad (6.4)$$

where only factors with $2^j \leq n$ affect the coefficient. Moreover, for $q, m \geq 0$,

$$\sum_{r=0}^{2^m-1} \varepsilon_{2^m q + r} z^r = \varepsilon_q P_m(z). \quad (6.5)$$

Proof. The first statement is coefficient stabilization in the finite products. The second is theorem 2.2 summed over r . □

Formal versions. The corpus proves the finite factorization as `Fabius.thueMorseBlockPolynomial_eq_product` (`ThueMorseGenerating.lean`); `ThueMorseBooleanCube.lean` now proves it over an *arbitrary* commutative ring as `Fabius.prod_one_sub_pow_eq_sum_thueMorseSign`, as the specialization $u = -1$ of a two-parameter identity

$$\prod_{j < m} (1 + u z^{2^j}) = \sum_{n < 2^m} u^{\text{wt}_2(n)} z^n \quad (\text{Fabius.prod_one_add_mul_pow}),$$

valid in any commutative semiring with no convergence hypothesis. Its engine is a reindexing kernel identifying the dyadic block with the powerset of $\{0, \dots, m-1\}$ (`Fabius.sum_power_set_two_pow`) together with the digit dictionary `Fabius.binaryWeight_sum_two_pow`, `Fabius.sum_two_pow_lt_two_pow`: exactly the Boolean-cube structure this section exploits informally.

6.3 Reciprocity and cyclotomic multiplicities

The reflection symmetry becomes a reciprocal-polynomial identity.

Proposition 6.4 (Reciprocity). *For every $m \geq 0$,*

$$z^{2^m-1} P_m(z^{-1}) = (-1)^m P_m(z). \quad (6.6)$$

Thus P_m is reciprocal for even m and anti-reciprocal for odd m .

Proof. Reverse the coefficients and apply (2.8); alternatively, apply $z^{2^j}(1 - z^{-2^j}) = z^{2^j} - 1 = -(1 - z^{2^j})$ to every factor. $\square$

Repeatedly factor $1 - z^{2^j}$ by difference of squares:

$$P_m(z) = (1 - z)^m \prod_{q=0}^{m-2} (1 + z^{2^q})^{m-1-q}. \quad (6.7)$$

If Φ_d is the d -th cyclotomic polynomial, then equivalently

$$P_m(z) = (-1)^m \prod_{q=0}^{m-1} \Phi_{2^q}(z)^{m-q}. \quad (6.8)$$

Here $\Phi_1(z) = z - 1$. In particular, a primitive 2^q -th root of unity is a zero of multiplicity $m - q$ when $0 \leq q < m$. This multiplicity table simultaneously explains Prouhet cancellation at $z = 1$ and the systematic zeros in the dyadic discrete Fourier transform.

Formal versions. `ThueMorseBlockAlgebra.lean` proves the reciprocity (6.6) in two forms: as the coefficient-word reversal `Fabius.reverse_thueMorseBlockPolynomial` ($P_m^{\text{rev}} = (-1)^m P_m$ through `Polynomial.reverse`) and as the evaluation identity `Fabius.sum_thueMorseSign_mul_pow_reflect` over an arbitrary commutative ring, which is (6.6) cleared of the denominator z^{2^m-1} . The degree data feeding it — $\deg P_m = 2^m - 1$, top and leading coefficient $(-1)^m$, $P_m \neq 0$ — are `Fabius.natDegree_thueMorseBlockPolynomial`, `Fabius.coeff_thueMorseBlockPolynomial_top`, `Fabius.leadingCoeff_thueMorseBlockPolynomial`, and `Fabius.thueMorseBlockPolynomial_ne_zero`. The difference-of-squares ladder (6.7) is `Fabius.prod_one_sub_two_pow_eq_ladder`, valid over any commutative ring and built on the telescope $(1 - z) \prod_{i < j} (1 + z^{2^i}) = 1 - z^{2^j}$ (`Fabius.one_sub_mul_prod_one_add_two_pow`); the resulting Prouhet divisibility $(1 - X)^m \mid P_m$ is `Fabius.one_sub_X_pow_dvd_thueMorseBlockPolynomial`. The cyclotomic table (6.8) is `Fabius.thueMorseBlockPolynomial_eq_cyclotomic_prod`, resting on $\Phi_{2^{q+1}} = 1 + z^{2^q}$ (`Fabius.cyclotomic_two_pow_succ`, over any commutative ring).

6.4 The infinite product and Mahler equation

For $|z| < 1$, define

$$E(z) = \sum_{n \geq 0} \varepsilon_n z^n, \quad T(z) = \sum_{n \geq 0} \tau(n) z^n. \quad (6.9)$$

Theorem 6.5 (Infinite lacunary product). *For $|z| < 1$, and coefficientwise as a formal power series,*

$$E(z) = \prod_{j=0}^{\infty} (1 - z^{2^j}). \quad (6.10)$$

The product converges uniformly on compact subsets of the unit disk. The zero–one series is

$$T(z) = \frac{1}{2(1-z)} - \frac{1}{2}E(z). \quad (6.11)$$

Proof. For $|z| \leq \rho < 1$, the sum $\sum_j |z|^{2^j}$ converges, so the product converges uniformly on the compact disk. Every coefficient stabilizes in the finite products P_m , where [theorem 6.2](#) identifies it as ε_n . The formula for T follows from $\tau(n) = (1 - \varepsilon_n)/2$. $\square$

Peeling off factors gives the Mahler and dissection equations.

Corollary 6.6 (Mahler equations and dissections). *For $|z| < 1$,*

$$E(z) = (1 - z)E(z^2), \quad (6.12)$$

$$E(z) = P_m(z)E(z^{2^m}), \quad (6.13)$$

$$T(z) = (1 - z)T(z^2) + \frac{z}{1 - z^2}. \quad (6.14)$$

Proof. Separate the first factor, or the first m factors, in (6.10). Substitute (6.11) for the final identity; it also follows directly from (2.2). $\square$

Iterating the inhomogeneous bit equation yields a useful lacunary sum.

Theorem 6.7 (Lacunary quotient expansion). *Let $P_0(z) = 1$ and $P_j(z) = \prod_{h=0}^{j-1} (1 - z^{2^h})$ for $j \geq 1$. Then, for $|z| < 1$,*

$$T(z) = \sum_{j=0}^{\infty} P_j(z) \frac{z^{2^j}}{1 - z^{2^{j+1}}}. \quad (6.15)$$

Proof. After M iterations of (6.14),

$$T(z) = \sum_{j=0}^{M-1} P_j(z) \frac{z^{2^j}}{1 - z^{2^{j+1}}} + P_M(z)T(z^{2^M}).$$

On compact subdisks P_M is bounded and $T(z^{2^M}) \rightarrow T(0) = 0$. $\square$

6.5 A weighted digit enumerator

Both E and the geometric series are specializations of one umbrella product that keeps the whole digit sum, not only its parity.

Theorem 6.8 (Weighted enumerator). *For $|z| < 1$ and any scalar u , and coefficientwise as formal series,*

$$F_u(z) := \sum_{n \geq 0} u^{\text{wt}_2(n)} z^n = \prod_{j \geq 0} (1 + uz^{2^j}), \quad (6.16)$$

with the Mahler equation

$$F_u(z) = (1 + uz)F_u(z^2). \quad (6.17)$$

The specializations $u = 1$ and $u = -1$ are

$$F_1(z) = \frac{1}{1 - z}, \quad F_{-1}(z) = E(z). \quad (6.18)$$

Proof. Expanding the product chooses a finite set of bit positions; by uniqueness of binary expansions each exponent n arises exactly once, with coefficient $u^{\text{wt}_2(n)}$. Peeling the first factor gives (6.17). At $u = 1$ every n receives coefficient one, which restates the uniqueness of binary expansions; at $u = -1$ the coefficient is ε_n . $\square$

Differentiating (6.16) with respect to u at $u = 1$ gives the ordinary generating function of the digit sum itself:

$$\sum_{n \geq 0} \text{wt}_2(n) z^n = \frac{1}{1 - z} \sum_{j \geq 0} \frac{z^{2^j}}{1 + z^{2^j}}. \quad (6.19)$$

Higher derivatives at $u = 1$ generate the falling-factorial moments $\text{wt}_2(n)(\text{wt}_2(n) - 1) \cdots (\text{wt}_2(n) - r + 1)$ in the same way. The lacunary sum in (6.19) is the characteristic-zero twin of the Artin–Schreier solution (7.16) below: the same series $\sum_j z^{2^j} / (1 + z^{2^j})$ computes the digit sum with integer coefficients and the digit parity in characteristic two. The umbrella product itself is formalized in full generality as `Fabius.prod_one_add_mul_pow (ThueMorseBooleanCube.lean)`; the specializations (6.18) are the $z = 1$ and $u = -1$ readings recorded there as `Fabius.sum_pow_bin`, `aryWeight_eq_one_add_pow` and `Fabius.prod_one_sub_pow_eq_sum_thueMorseSign`.

6.6 Thue–Morse constants in every base

For a real $b > 1$, put

$$\Theta_b = \sum_{n \geq 0} \frac{\tau(n)}{b^{n+1}}. \quad (6.20)$$

For integer $b \geq 2$, these are the base- b numbers whose fractional digits are the Thue–Morse bits.

Corollary 6.9 (Base- b product). *For every real $b > 1$,*

$$\Theta_b = \frac{1}{2(b-1)} - \frac{1}{2b} \prod_{j \geq 0} (1 - b^{-2^j}). \quad (6.21)$$

The signed companion is

$$\sum_{n \geq 0} \frac{\varepsilon_n}{b^{n+1}} = \frac{1}{b} \prod_{j \geq 0} (1 - b^{-2^j}). \quad (6.22)$$

Proof. Set $z = 1/b$ in (6.11) and (6.10), then multiply by $1/b$. $\square$

For $b = 2$, $\Theta_2 = 0.0110100110010110 \dots_2$ is the classical Prouhet–Thue–Morse constant. Mahler’s theory implies its transcendence; that deep arithmetic result is separate from the elementary product identity proved here [8, 12].

7 Euler transforms, coefficient recurrences, and algebraic series

7.1 The logarithmic ruler transform

Take the formal logarithm of (6.10):

$$\log E(z) = \sum_{j \geq 0} \log(1 - z^{2^j}) = - \sum_{j \geq 0} \sum_{r \geq 1} \frac{z^{r2^j}}{r}. \quad (7.1)$$

For a fixed exponent k , the admissible j 's satisfy $0 \leq j \leq \nu_2(k)$, and the corresponding denominator is $k/2^j$. Define

$$a_k = \sum_{j=0}^{\nu_2(k)} 2^j = 2^{\nu_2(k)+1} - 1. \quad (7.2)$$

Theorem 7.1 (Euler transform of the ruler function). *As a formal series, and analytically for $|z| < 1$,*

$$\log E(z) = - \sum_{k \geq 1} \frac{a_k}{k} z^k, \quad a_k = 2^{\nu_2(k)+1} - 1. \quad (7.3)$$

This representation contains no explicit binary digit sum. The coefficient sequence (a_k) is a simple transform of the ruler sequence.

7.2 A nonlocal convolution recurrence

Theorem 7.2 (Ruler-convolution recurrence). *With $\varepsilon_0 = 1$, for every $n \geq 1$,*

$$n\varepsilon_n = - \sum_{k=1}^n a_k \varepsilon_{n-k}. \quad (7.4)$$

Proof. Differentiate (7.3), multiply by $E(z)$, and compare the coefficient of z^{n-1} :

$$E'(z) = -E(z) \sum_{k \geq 1} a_k z^{k-1}.$$

□

The weighted convolution on the right always collapses to $\pm n$. As an evaluator it is inferior to popcount, but it is the natural recurrence attached to the Euler product.

7.3 Partition and Bell-polynomial formulae

Exponentiating (7.3) gives a finite partition sum for every coefficient.

Theorem 7.3 (Partition formula). *For $n \geq 0$,*

$$\varepsilon_n = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \prod_{j=1}^n \frac{1}{m_j!} \left(-\frac{a_j}{j} \right)^{m_j}. \quad (7.5)$$

Proof. Write $E(z) = \prod_{j \geq 1} \exp(-a_j z^j / j)$, expand each exponential, and collect terms of total degree n . □

Let $\mathcal{B}_n$ denote the complete exponential Bell polynomial, defined by

$$\exp\left(\sum_{j \geq 1} x_j \frac{z^j}{j!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (7.6)$$

Corollary 7.4 (Bell-polynomial coefficient). *For every $n \geq 0$,*

$$\varepsilon_n = \frac{1}{n!} \mathcal{B}_n(-0!a_1, -1!a_2, \dots, -(n-1)!a_n). \quad (7.7)$$

Proof. Use $x_j = -(j-1)!a_j$ in (7.6). □

Although the right sides are rational-looking sums, they always equal ± 1 . This integrality collapse is itself encoded by the original lacunary product.

7.4 A Hessenberg determinant

For $n \geq 1$, let

$$H_n = \begin{pmatrix} a_1 & 1 & 0 & \cdots & 0 \\ a_2 & a_1 & 2 & \ddots & \vdots \\ a_3 & a_2 & a_1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & n-1 \\ a_n & a_{n-1} & \cdots & a_2 & a_1 \end{pmatrix}. \quad (7.8)$$

Theorem 7.5 (Determinant formula). *For every $n \geq 1$,*

$$\varepsilon_n = \frac{(-1)^n}{n!} \det H_n. \quad (7.9)$$

In particular,

$$\det H_n = (-1)^n n! \varepsilon_n \in \{n!, -n!\}. \quad (7.10)$$

Proof. Put $d_0 = 1$ and $d_n = \det H_n$. Expansion of a lower Hessenberg determinant along its first column gives

$$d_n = \sum_{k=1}^n (-1)^{k-1} \frac{(n-1)!}{(n-k)!} a_k d_{n-k}. \quad (7.11)$$

Set $d_n = (-1)^n n! c_n$. Then (7.11) reduces to

$$n c_n = - \sum_{k=1}^n a_k c_{n-k},$$

with $c_0 = 1$. By uniqueness of the recurrence in [theorem 7.2](#), $c_n = \varepsilon_n$. □

7.5 The zero–one series over $\mathbb{F}_2$

The signed series loses all information modulo two because $-1 = 1$. The bit series remains nontrivial. Work in $\mathbb{F}_2[[z]]$ and write

$$\Theta(z) = \sum_{n \geq 0} \tau(n) z^n. \quad (7.12)$$

Theorem 7.6 (Artin–Schreier equation). *In $\mathbb{F}_2[[z]]$,*

$$\Theta(z) = (1+z)\Theta(z^2) + \frac{z}{1+z^2}, \quad (7.13)$$

$$\boxed{(1+z)^3\Theta(z)^2 + (1+z)^2\Theta(z) + z = 0.} \quad (7.14)$$

If $Y(z) = (1+z)\Theta(z)$, then

$$\boxed{Y(z)^2 + Y(z) = \frac{z}{1+z}.} \quad (7.15)$$

Proof. Split the series into even and odd coefficients and use $\tau(2n) = \tau(n)$, $\tau(2n+1) = 1 + \tau(n)$ in characteristic two:

$$\Theta(z) = \Theta(z^2) + z \sum_{n \geq 0} (1 + \tau(n)) z^{2n} = (1+z)\Theta(z^2) + \frac{z}{1+z^2}.$$

Frobenius gives $\Theta(z^2) = \Theta(z)^2$, while $1 + z^2 = (1+z)^2$. Clear the denominator and substitute $Y = (1+z)\Theta$. $\square$

Corollary 7.7 (Explicit Artin–Schreier sum). *The unique zero-constant-term solution is*

$$Y(z) = \sum_{j=0}^{\infty} \left(\frac{z}{1+z} \right)^{2^j} = \sum_{j=0}^{\infty} \frac{z^{2^j}}{1+z^{2^j}}, \quad (7.16)$$

$$\boxed{\Theta(z) = \frac{1}{1+z} \sum_{j=0}^{\infty} \frac{z^{2^j}}{1+z^{2^j}.} \quad (7.17)$$

Proof. Let $A = z/(1+z)$. In the z -adic topology, $Y = \sum_{j \geq 0} A^{2^j}$ is well defined and $Y^2 + Y = A$ by index shifting. Frobenius gives $A^{2^j} = z^{2^j}/(1+z^{2^j})$. A zero-constant solution of the homogeneous equation $Z^2 + Z = 0$ must vanish, proving uniqueness. $\square$

Christol’s theorem says that a finite-field sequence is automatic exactly when its generating series is algebraic. Here the degree-two certificate is derived directly; it is an algebraic shadow of the two-state automaton [13].

7.6 An integral algebraic lift

The fourth draft records a rather different integer sequence whose parity is Thue–Morse. Define $c_0 = 0$, $c_1 = 1$, and for $n \geq 2$,

$$nc_n = (5 - 2n)c_{n-1} + 3(n-1)c_{n-2} + 1. \quad (7.18)$$

Its first terms are

$$0, 1, 1, 2, 1, 4, -2, 13, -23, 68, -164, 439, -1146, 3067, \dots$$

Theorem 7.8 (Integer algebraic lift). *Let $C(z) = \sum_{n \geq 0} c_n z^n$. Then*

$$C(z) = \frac{1}{2(1-z)} \left(\sqrt{\frac{1+3z}{1-z}} - 1 \right), \quad (7.19)$$

$$(1-z)^3 C(z)^2 + (1-z)^2 C(z) = z. \quad (7.20)$$

All c_n are integers and

$$\boxed{c_n \equiv \tau(n) \pmod{2}, \quad \varepsilon_n = (-1)^{c_n}.} \quad (7.21)$$

Proof. Multiplying (7.18) by z^{n-1} , summing, and including the initial terms gives a first-order differential equation whose zero-constant solution is (7.19); squaring yields (7.20).

Conversely, the coefficient of z^n in $(1-z)^2 C(z)$ contains c_n with coefficient one, while the quadratic term uses only earlier coefficients because $C(0) = 0$. Thus (7.20) recursively determines integers c_n .

Modulo two, (7.20) becomes precisely (7.14). That equation has a unique zero-constant solution in $\mathbb{F}_2[[z]]$, so $C(z) \equiv \Theta(z) \pmod{2}$. $\square$

Remark 7.9. The lift grows and changes sign; it is not a disguised digit sum. Its parity is Thue–Morse because an integral algebraic equation reduces to the automatic sequence’s Artin–Schreier equation. This suggests a reusable discovery method: search for integral algebraic lifts of finite-field automatic series.

8 Natural boundary, Hankel determinants, and Diophantine consequences

The algebraic picture of the previous section is completed by three analytic-arithmetic facts: the series E cannot be continued across any boundary point, its Hankel determinants never vanish, and its values at algebraic points are transcendental with the smallest possible irrationality exponent. The first fact is proved here in full; the other two are deep literature theorems, stated with references and not reproved.

8.1 The unit circle is a natural boundary

A *natural boundary* of a power series is a curve through which no analytic continuation across any arc is possible.

Theorem 8.1 (Natural boundary). *Every point of the unit circle is a singular point of E ; the circle is a natural boundary. Consequently E and T are transcendental over the field $\mathbb{C}(z)$ of rational functions.*

Proof. Let ζ be a root of unity of two-power order, say $\zeta^{2^r} = 1$. For $0 < \rho < 1$, split the product (6.10) at index r : the head has r factors, each of modulus at most 2, while every tail factor $1 - (\rho\zeta)^{2^j} = 1 - \rho^{2^j}$ with $j \geq r$ is a real number in $(0, 1)$. Hence

$$|E(\rho\zeta)| \leq 2^r (1 - \rho^{2^r}) \xrightarrow{\rho \uparrow 1} 0.$$

Such roots of unity are dense on the circle. If E continued analytically through some open boundary arc, the continued function would vanish on a set with an accumulation point inside its domain, hence would vanish identically by the identity theorem, contradicting $E(0) = 1$. An algebraic function over $\mathbb{C}(z)$ has only finitely many singular points on the Riemann sphere, so it cannot have a natural boundary; and T differs from $-E/2$ by the rational function in (6.11). $\square$

Remark 8.2. Two softer routes to the same conclusion deserve mention. The Pólya–Carlson theorem [20] says that a power series with integer coefficients and radius of convergence one is either rational or has the unit circle as a natural boundary, and E is visibly not rational because its coefficient sequence is not eventually periodic (section 14). For Mahler functions there is also a general rational-or-natural-boundary dichotomy [21]. The radial-zero argument above is preferred here because it is self-contained.

The contrast with section 7.5 is worth displaying:

$$\left. \begin{array}{l} \text{over } \mathbb{F}_2(z) \\ \text{over } \mathbb{C}(z) \end{array} \right| \begin{array}{l} \Theta(z) \text{ is algebraic of degree two,} \\ T(z) \text{ is transcendental with a natural boundary.} \end{array} \quad (8.1)$$

Automaticity is algebraicity in the matching finite characteristic, not in characteristic zero.

8.2 Hankel determinants never vanish

For a sequence $c = (c_n)_{n \geq 0}$, the order- r *Hankel determinant* is the determinant of the $r \times r$ matrix whose (i, j) -entry is c_{i+j} :

$$H_r(c) = \det(c_{i+j})_{0 \leq i, j < r}. \quad (8.2)$$

For the signed sequence the first values are

$$H_1, H_2, \dots = 1, -2, 4, 8, -16, -32, -64, 128, -256, -1536, \dots \quad (8.3)$$

Unlike the Hessenberg matrix of [theorem 7.5](#), which was engineered from the ruler weights precisely so that its determinant would telescope, these matrices sample the raw signs and have no visible triangular structure.

Theorem 8.3 (Allouche–Peyrière–Wen–Wen; literature theorem). *For every $r \geq 1$,*

$$\boxed{H_r(\varepsilon) \neq 0.} \quad (8.4)$$

The original proof runs a closed finite system of determinant recurrences [22]; later proofs include Han’s Hankel continued-fraction calculus [23] and a combinatorial argument of Bugeaud and Han [24]. The statement is a strong rigidity property: no finite window of the sequence ever satisfies an accidental low-order linear relation of Hankel type.

Nonvanishing has a standard approximation-theoretic reading. A *Padé approximant of type $[r - 1/r]$* to E is a rational function $A(z)/B(z)$ with $\deg A \leq r - 1$ and $\deg B \leq r$ whose Taylor expansion agrees with E to the maximal possible order. When $H_r(\varepsilon) \neq 0$, this approximant exists and the agreement is exact through order z^{2r-1} with a nonzero discrepancy coefficient at z^{2r} ; the whole diagonal Padé table of E is therefore normal, with no degenerate blocks.

8.3 Transcendental values and the sharp irrationality exponent

Beyond transcendence of the function, Mahler’s method gives transcendence of its values. The Mahler equation (6.12) relates z to z^2 , and for an algebraic α with $0 < |\alpha| < 1$ the forward orbit $\alpha, \alpha^2, \alpha^4, \dots$ stays in the unit disk and never meets the zero of the rational cofactor $1 - z$.

Theorem 8.4 (Mahler; literature theorem). *If α is algebraic and $0 < |\alpha| < 1$, then $E(\alpha)$ is transcendental; consequently so is $T(\alpha)$ [8].*

In particular, every base- b Thue–Morse constant Θ_b of (6.20) is transcendental for integer $b \geq 2$. How well such a constant can be approximated by rationals is measured by its *irrationality exponent* $\mu(\xi)$: the supremum of all real μ such that $|\xi - p/q| < q^{-\mu}$ holds for infinitely many fractions p/q . Every irrational number has $\mu \geq 2$, by Dirichlet’s pigeonhole theorem, and Lebesgue-almost every real number has $\mu = 2$.

Theorem 8.5 (Bugeaud; literature theorem). *For every integer $b \geq 2$,*

$$\boxed{\mu(\Theta_b) = 2.} \quad (8.5)$$

Thus the Thue–Morse–Mahler numbers are transcendental yet approximate as badly as possible for irrational numbers. Bugeaud’s proof [25] runs the Padé approximants of the previous subsection through the Mahler equation: iterating $z \mapsto z^2$ transports one good rational approximation at $z = 1/b$ to a whole geometric family, and the nonvanishing Hankel determinants guarantee that the approximation orders have no long gaps. Earlier partial bounds and the general determinant framework are surveyed in [26, 27]. This is the cleanest place where the determinant rigidity of [theorem 8.3](#) pays an explicitly Diophantine dividend.

9 Finite differences, Prouhet cancellation, and all dyadic moments

9.1 The mixed-difference master identity

For a function f , let

$$(\Delta_h f)(x) = f(x+h) - f(x), \quad (T_h f)(x) = f(x+h). \quad (9.1)$$

Thus $I - T_h = -\Delta_h$.

Theorem 9.1 (Dyadic finite-difference factorization). *For every $m \geq 0$, every scalar h , and every function for which the values are defined,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n f(x+nh) = \prod_{j=0}^{m-1} (I - T_{2^j h}) f(x) \quad (9.2)$$

$$= (-1)^m (\Delta_h \Delta_{2h} \cdots \Delta_{2^{m-1}h} f)(x). \quad (9.3)$$

Proof. Expand the operator product. A subset $S \subseteq \{0, \dots, m-1\}$ contributes the shift by $h \sum_{j \in S} 2^j$ and sign $(-1)^{|S|}$. Binary expansion identifies the subset sum with a unique $n < 2^m$, and the sign is ε_n . $\square$

This theorem is stronger than any one moment identity: it is an inclusion–exclusion formula valid for arbitrary test functions.

9.2 Exact multiple-integral remainder

Corollary 9.2 (Integral cubature formula). *Suppose $h \geq 0$ and $f \in C^m$ on an interval containing all relevant points. Then*

$$\sum_{n < 2^m} \varepsilon_n f(x+nh) = (-1)^m \int_0^h \int_0^{2h} \cdots \int_0^{2^{m-1}h} f^{(m)}(x+t_0+\cdots+t_{m-1}) dt_{m-1} \cdots dt_0 \quad (9.4)$$

$$= (-1)^m 2^{\binom{m}{2}} h^m \int_{[0,1]^m} f^{(m)}\left(x+h \sum_{j=0}^{m-1} 2^j u_j\right) d\mathbf{u}. \quad (9.5)$$

Consequently

$$\left| \sum_{n < 2^m} \varepsilon_n f(x+nh) \right| \leq 2^{\binom{m}{2}} h^m \sup_{x \leq y \leq x+(2^m-1)h} |f^{(m)}(y)|. \quad (9.6)$$

Proof. Apply $(I - T_a)f(x) = -\int_0^a f'(x+t) dt$ successively. Rescale $t_j = 2^j h u_j$ for the unit-cube form. The box volume is $h^m 2^{0+1+\cdots+(m-1)}$. $\square$

Corollary 9.3 (Sign for completely monotone kernels). *If $(-1)^m f^{(m)} \geq 0$ on the interval, then*

$$\sum_{n < 2^m} \varepsilon_n f(x+nh) \geq 0. \quad (9.7)$$

For $a, s > 0$,

$$\sum_{n < 2^m} \frac{\varepsilon_n}{(a+n)^s} = (s)_m \int_0^1 \int_0^2 \cdots \int_0^{2^{m-1}} \frac{dt_{m-1} \cdots dt_0}{(a+t_0+\cdots+t_{m-1})^{s+m}} > 0, \quad (9.8)$$

where $(s)_m = s(s+1) \cdots (s+m-1)$. For $m \geq 1$,

$$\sum_{n < 2^m} \varepsilon_n \log(a+n) < 0, \quad 0 < \prod_{n < 2^m} (a+n)^{\varepsilon_n} < 1. \quad (9.9)$$

Proof. Use $f(y) = (a+y)^{-s}$, for which $f^{(m)}(y) = (-1)^m (s)_m (a+y)^{-s-m}$. For the logarithm, $f^{(m)}(y) = (-1)^{m-1} (m-1)! / (a+y)^m$, so the integral is strictly negative. Exponentiation gives the product inequality. $\square$

9.3 Arbitrary translation weights

The dyadic steps $h, 2h, 4h, \dots$ give the identities above their Thue–Morse reading, but the cancellation mechanism does not use them. The formalization isolates the general statement, which is worth recording: it subsumes the ordinary and sparse Prouhet identities simultaneously and explains exactly where the powers of two matter (only in the value of the sharp moment, never in the vanishing).

Theorem 9.4 (Arbitrary-weight Prouhet identities). *Let R be a commutative ring, S a finite index set, $w: S \rightarrow R$ arbitrary weights, and $x \in R$. For every polynomial $p \in R[y]$ with $\deg p < |S|$,*

$$\sum_{T \subseteq S} (-1)^{|T|} p\left(x + \sum_{j \in T} w_j\right) = 0, \quad (9.10)$$

and at the first surviving degree,

$$\boxed{\sum_{T \subseteq S} (-1)^{|T|} \left(x + \sum_{j \in T} w_j\right)^{|S|} = (-1)^{|S|} |S|! \prod_{j \in S} w_j.} \quad (9.11)$$

Proof. For (9.10), induct on S . If $S = \emptyset$ there is nothing of degree below zero. For $S \cup \{a\}$ with $a \notin S$, split the subsets by membership of a and pair T with $T \cup \{a\}$:

$$\sum_{T \subseteq S \cup \{a\}} (-1)^{|T|} p(x + w_T) = \sum_{T \subseteq S} (-1)^{|T|} q(x + w_T), \quad q(y) := p(y) - p(y + w_a),$$

where $w_T := \sum_{j \in T} w_j$. The translation $p(y + w_a)$ has the same degree and the same leading coefficient as p , so $\deg q < \deg p$ (and $q = 0$ when p is constant). The induction hypothesis applied to q finishes.

For (9.11), induct on S again with $p(y) = y^{s+1}$, $s = |S|$. The same pairing produces $q(y) = y^{s+1} - (y + w_a)^{s+1} = -\sum_{k \leq s} \binom{s+1}{k} w_a^{s+1-k} y^k$ by the binomial theorem, the top term cancelling. Summing over $T \subseteq S$ and exchanging the two finite sums, every row $k < s$ vanishes by (9.10), and the row $k = s$ contributes $-\binom{s+1}{s} w_a \cdot (-1)^s s! \prod_{j \in S} w_j$ by the induction hypothesis. Since $\binom{s+1}{s} = s+1$, the total is $(-1)^{s+1} (s+1)! w_a \prod_{j \in S} w_j$, as claimed. $\square$

Formal versions. `Fabius.sum_powerset_neg_one_pow_eval` and `Fabius.sum_powerset_neg_one_pow_pow_card`, in `ThueMorseSparseProuhet.lean`; the degree drop is `Mathlib's Polynomial.taylor` bookkeeping. Specializing $w_j = 2^j h$ over $S = \{0, \dots, m-1\}$ and transporting along the Boolean-cube kernel yields the Thue–Morse forms over any commutative ring: `Fabius.sum_thueMorseSign_mul_affine_pow_eq_zero` and `Fabius.sum_thueMorseSign_mul_affine_pow_card`, the latter with the familiar constant $\prod_{j < m} 2^j h = 2^{\binom{m}{2}} h^m$.

9.4 Prouhet cancellation and the first surviving moment

Theorem 9.5 (Prouhet polynomial annihilation). *If p is a polynomial of degree less than m , then*

$$\sum_{n=0}^{2^m-1} \varepsilon_n p(x + nh) = 0. \quad (9.12)$$

If $p(y) = cy^m + \text{lower-degree terms}$, then

$$\sum_{n=0}^{2^m-1} \varepsilon_n p(x + nh) = (-1)^m m! 2^{\binom{m}{2}} h^m c. \quad (9.13)$$

Proof. Every m -fold finite difference annihilates a polynomial of degree below m . At degree exactly m , its m -th derivative is the constant $m!c$, so (9.5) gives the sharp value. $\square$

For $h = 1$, let

$$A_m = \{0 \leq n < 2^m : \tau(n) = 0\}, \quad B_m = \{0 \leq n < 2^m : \tau(n) = 1\}. \quad (9.14)$$

Then

$$\sum_{n \in A_m} n^r = \sum_{n \in B_m} n^r, \quad 0 \leq r < m, \quad (9.15)$$

which is Prouhet’s classical equal-sums-of-like-powers construction. At $r = m$, the difference is nonzero, so the cancellation order is exact.

Remark 9.6 (The Prouhet–Tarry–Escott problem). Equation (9.15) is the historical origin of the *Prouhet–Tarry–Escott problem*: given a degree d , find two disjoint multisets $A \neq B$ of integers of equal size with $\sum_{a \in A} a^r = \sum_{b \in B} b^r$ for all $0 \leq r \leq d$. A solution is called *ideal* when its size is $d + 1$, the smallest conceivable. Prouhet’s partition (9.14) solves degree $d = m - 1$ with the much larger size 2^{m-1} , but it does so uniformly for every degree – still the only completely elementary construction that scales [2, 17, 18]. Ideal solutions are known for all $d \leq 9$ and for $d = 11$; their existence for every degree is open. The sparse Prouhet theorem of section 10 produces further nonconsecutive solutions: for any n with $\text{wt}_2(n) = s$, splitting the submasks of n by sign parity gives two sets of size 2^{s-1} with equal power sums through degree $s - 1$, supported on the odd positions of Pascal row n rather than on an interval.

9.5 The finite exponential transform

Define

$$M_{m,r} = \sum_{n=0}^{2^m-1} \varepsilon_n n^r. \quad (9.16)$$

Substitute $z = e^t$ in (6.3):

$$\sum_{r \geq 0} M_{m,r} \frac{t^r}{r!} = \sum_{n < 2^m} \varepsilon_n e^{nt} = \prod_{j=0}^{m-1} (1 - e^{2^j t}). \quad (9.17)$$

The order- m zero at $t = 0$ immediately gives

$$M_{m,r} = 0 \quad (r < m), \quad M_{m,m} = (-1)^m m! 2^{\binom{m}{2}}. \quad (9.18)$$

Formal versions. The vanishing and sharp value (9.18) have long been formal – for an *arbitrary* rational translation – as `Fabius.thueMorseTranslatedPowerSum_eq_zero_of_lt` and `Fabius.thueMorseTranslatedPowerSum_self` (`ThueMorseExponential.lean`); the raw moments $M_{m,r}$ are the translation $c = 2^m$ of that interface, `Fabius.thueMorseTranslatedPowerSum_two_pow` (`ThueMorseMoments.lean`). The transform (9.17) itself is formal at the coefficient level: the dictionary $[t^r] \sum_{n < 2^m} \varepsilon_n e^{nt} = M_{m,r}/r!$ is `Fabius.coeff_sum_thueMorseSign_exp_pow`, and the right-hand product is the algebraic master product evaluated at $z = e^t \in \mathbb{Q}[[t]]$ – no convergence argument is used anywhere.

9.6 All power moments: a finite composition formula

Theorem 9.7 (Complete moment composition). *For $m \geq 1$ and $r \geq m$,*

$$M_{m,r} = (-1)^m r! 2^{\binom{m}{2}} \sum_{\substack{q_0, \dots, q_{m-1} \geq 0 \\ q_0 + \dots + q_{m-1} = r-m}} \prod_{j=0}^{m-1} \frac{2^{jq_j}}{(q_j + 1)!}. \quad (9.19)$$

Equivalently,

$$M_{m,r} = (-1)^m r! \sum_{\substack{p_0, \dots, p_{m-1} \geq 1 \\ p_0 + \dots + p_{m-1} = r}} \frac{2^{\sum_{j=0}^{m-1} jp_j}}{p_0! \cdots p_{m-1}!}. \quad (9.20)$$

Proof. Factor the lowest-order term from each factor in (9.17):

$$\begin{aligned} \prod_{j=0}^{m-1} (1 - e^{2^j t}) &= (-1)^m 2^{\binom{m}{2}} t^m \prod_{j=0}^{m-1} \frac{e^{2^j t} - 1}{2^j t} \\ &= (-1)^m 2^{\binom{m}{2}} t^m \prod_{j=0}^{m-1} \sum_{q \geq 0} \frac{2^{jq} t^q}{(q+1)!}. \end{aligned}$$

Extract t^r and multiply by $r!$. Set $p_j = q_j + 1$ for the second form. $\square$

Formal versions. `ThueMorseMoments.lean` proves [theorem 9.7](#) as `Fabius.sum_thueMorseSign_mul_pow_add`: for all $d \geq 0$,

$$\sum_{n < 2^m} \varepsilon_n n^{m+d} = (-1)^m (m+d)! \sum_{\substack{q_0, \dots, q_{m-1} \geq 0 \\ q_0 + \dots + q_{m-1} = d}} \prod_{j=0}^{m-1} \frac{(2^j)^{q_j+1}}{(q_j + 1)!},$$

with the compositions realized as finitely supported functions (`Finset.finsuppAntidiag`). Factoring $(2^j)^{q_j+1} = 2^j \cdot 2^{jq_j}$ across the product recovers the boxed constant $2^{\binom{m}{2}}$ of (9.19), and $p_j = q_j + 1$ gives (9.20). The formal proof substitutes $z = e^t$ into the algebraic master product – each factor becomes $-t$ times an explicit entire series (`Fabius.one_sub_exp_two_pow`, `Fabius.prod_one_sub_exp_two_pow`) – and extracts the t^{m+d} coefficient through `PowerSeries.coeff_prod`.

9.7 Bell-polynomial compression and cumulants

Let $B_{2\ell}$ be the Bernoulli numbers with $B_2 = 1/6$, and define

$$\kappa_1^{(m)} = \frac{2^m - 1}{2}, \quad (9.21)$$

$$\kappa_{2\ell}^{(m)} = \frac{B_{2\ell}}{2\ell} \frac{2^{2\ell m} - 1}{2^{2\ell} - 1} \quad (\ell \geq 1), \quad (9.22)$$

$$\kappa_{2\ell+1}^{(m)} = 0 \quad (\ell \geq 1). \quad (9.23)$$

Theorem 9.8 (Bell-polynomial moments). *For $m \geq 1$ and $r \geq 0$,*

$$M_{m,m+r} = (-1)^m 2^{\binom{m}{2}} \frac{(m+r)!}{r!} \mathcal{B}_r \left(\kappa_1^{(m)}, \dots, \kappa_r^{(m)} \right). \quad (9.24)$$

Proof. The classical formal expansion

$$\log \left(\frac{e^y - 1}{y} \right) = \frac{y}{2} + \sum_{\ell \geq 1} \frac{B_{2\ell}}{2\ell(2\ell)!} y^{2\ell} \quad (9.25)$$

turns the normalized product into

$$\prod_{j=0}^{m-1} (1 - e^{2^j t}) = (-1)^m 2^{\binom{m}{2}} t^m \exp \left(\sum_{r \geq 1} \kappa_r^{(m)} \frac{t^r}{r!} \right). \quad (9.26)$$

Use the defining generating function (7.6) and compare coefficients. $\square$

The first three nonzero layers are

$$M_{m,m} = (-1)^m 2^{\binom{m}{2}} m!, \quad (9.27)$$

$$M_{m,m+1} = (-1)^m 2^{\binom{m}{2}} (m+1)! \frac{2^m - 1}{2}, \quad (9.28)$$

$$M_{m,m+2} = (-1)^m 2^{\binom{m}{2}} (m+2)! \left(\frac{(2^m - 1)^2}{8} + \frac{4^m - 1}{72} \right). \quad (9.29)$$

9.8 Centered moments and a parity selection rule

Let

$$c_m = \frac{2^m - 1}{2}. \quad (9.30)$$

The centered exponential transform is especially symmetric.

Theorem 9.9 (Centered sinh product). *For every $m \geq 0$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n e^{(n-c_m)t} = (-2)^m \prod_{j=0}^{m-1} \sinh(2^{j-1}t). \quad (9.31)$$

Consequently

$$\sum_{n < 2^m} \varepsilon_n (n - c_m)^r = 0 \quad (9.32)$$

whenever $r < m$ or $r \not\equiv m \pmod{2}$. At $r = m$,

$$\sum_{n < 2^m} \varepsilon_n (n - c_m)^m = (-1)^m m! 2^{\binom{m}{2}}. \quad (9.33)$$

Proof. Use $1 - e^u = -2e^{u/2} \sinh(u/2)$ in (9.17). The product of exponential factors is $e^{c_m t}$, which disappears after centering. The right side is a product of m odd functions, hence has parity m , and each factor starts with a linear term. Coefficient comparison proves the selection rule and the first coefficient. $\square$

Remark 9.10 (Reflection proof of the selection rule). The parity cancellation (9.32) needs no exponential transform. Reindex $n \mapsto 2^m - 1 - n$ in $S_r = \sum_{n < 2^m} \varepsilon_n (n - c_m)^r$: the reflection formula (2.8) gives $\varepsilon_{2^m-1-n} = (-1)^m \varepsilon_n$, while $(2^m - 1 - n) - c_m = -(n - c_m)$. Hence $S_r = (-1)^{m+r} S_r$, and $S_r = 0$ whenever $m + r$ is odd. The same reindexing at an *arbitrary* translation x yields the functional equation

$$\sum_{n < 2^m} \varepsilon_n (x + n)^r = (-1)^{m+r} \sum_{n < 2^m} \varepsilon_n (n - x - (2^m - 1))^r,$$

valid over any commutative ring; the selection rule is its fixed point at the self-reflective translation $x = -c_m$.

Formal versions. The reflection functional equation is `Fabius.sum_thueMorseSign_mul_add_pow_reflect` (`ThueMorseMoments.lean`, any commutative ring); the selection rule (9.32) for odd $m + r$ is `Fabius.sum_thueMorseSign_mul_midpoint_pow_eq_zero`, proved by the reflection argument of [theorem 9.10](#) rather than through (9.31); the complementary case $r < m$ of either parity is `Fabius.thueMorseTranslatedPowerSum_eq_zero_of_lt`. The first coefficient (9.33) is `Fabius.sum_thueMorseSign_mul_midpoint_pow_self`.

This extra parity cancellation is invisible in the uncentered Prouhet statement. It is a consequence of combining dyadic reflection with moment cancellation.

9.9 All binomial-basis moments

Define

$$B_{m,r} = \sum_{n=0}^{2^m-1} \varepsilon_n \binom{n}{r}. \quad (9.34)$$

Theorem 9.11 (Binomial moment product). *For $\ell \geq 0$,*

$$B_{m,m+\ell} = (-1)^m 2^{\binom{m}{2}} [u^\ell] \prod_{j=0}^{m-1} \frac{(1+u)^{2^j} - 1}{2^j u}. \quad (9.35)$$

For $r < m$, $B_{m,r} = 0$; in particular,

$$B_{m,m} = (-1)^m 2^{\binom{m}{2}}. \quad (9.36)$$

Proof. Expand $P_m(1+u)$ in two ways:

$$P_m(1+u) = \sum_{n < 2^m} \varepsilon_n (1+u)^n = \sum_{r \geq 0} B_{m,r} u^r,$$

while

$$P_m(1+u) = (-1)^m 2^{\binom{m}{2}} u^m \prod_{j=0}^{m-1} \frac{(1+u)^{2^j} - 1}{2^j u}.$$

Compare coefficients. □

Corollary 9.12 (Integer composition form). *For all $m, \ell \geq 0$,*

$$B_{m,m+\ell} = (-1)^m \sum_{\substack{q_0, \dots, q_{m-1} \geq 0 \\ q_0 + \dots + q_{m-1} = \ell}} \prod_{j=0}^{m-1} \binom{2^j}{q_j + 1}. \quad (9.37)$$

Proof. In (9.35) each normalized factor is, up to its constant 2^j , the geometric sum $((1+u)^{2^j} - 1)/u = \sum_{i < 2^j} (1+u)^i$, whose q -th coefficient is $\sum_{i < 2^j} \binom{i}{q} = \binom{2^j}{q+1}$ by the hockey-stick identity. Multiplying the m factors and collecting the coefficient of u^ℓ gives (9.37). The single composition at $\ell = 0$ recovers (9.36), since $\binom{2^j}{1} = 2^j$ multiplies to $2^{\binom{m}{2}}$. Unlike the power-moment composition (9.19), every summand here is an integer. $\square$

Formal versions. `ThueMorseBlockAlgebra.lean` proves the substitution identity behind (9.35) as `Fabius.sum_thueMorseSign_mul_one_add_X_pow` (over any commutative ring), the vanishing $B_{m,r} = 0$ for $r < m$ as `Fabius.sum_thueMorseSign_mul_choose_eq_zero`, and the sharp value (9.36) as `Fabius.sum_thueMorseSign_mul_choose_self`, each with an integer specialization. The complete composition form (9.37) is `Fabius.sum_thueMorseSign_mul_choose_add` (`ThueMorseMoments.lean`), with the hockey-stick column sum $\sum_{i < N} \binom{i}{q} = \binom{N}{q+1}$ isolated as the reusable `Fabius.sum_range_choose_eq_choose_succ`.

The binomial basis is often preferable in discrete formalization because finite differences obey the simple lowering rule $\Delta \binom{x}{r} = \binom{x}{r-1}$.

10 Sparse Prouhet identities on arbitrary Pascal supports

The ordinary Prouhet block $[0, 2^m)$ is the submask lattice of $2^m - 1$. Lucas’s theorem suggests replacing this all-ones mask by an arbitrary n .

Define

$$J(n) = \{j \geq 0 : b_j(n) = 1\}, \quad s = |J(n)| = \text{wt}_2(n), \quad \beta(n) = \sum_{j \in J(n)} j. \quad (10.1)$$

Theorem 10.1 (Signed Pascal-support product). *For every $n \geq 0$,*

$$R_n(z) := \sum_{k \preceq_2 n} \varepsilon_k z^k = \prod_{j \in J(n)} (1 - z^{2^j}). \quad (10.2)$$

Equivalently,

$$R_n(z) = \sum_{k=0}^n \varepsilon_k \mathbf{1}_{\binom{n}{k} \text{ odd}} z^k. \quad (10.3)$$

Proof. Expanding the product chooses a subset of the occupied bit positions of n . Its exponent is the corresponding submask k , and its sign is $(-1)^{\text{wt}_2(k)} = \varepsilon_k$. Lucas’s criterion gives the second form. $\square$

Corollary 10.2 (Balance on odd Pascal positions). *For every $n > 0$,*

$$\sum_{\substack{0 \leq k \leq n \\ \binom{n}{k} \text{ odd}}} \varepsilon_k = 0. \quad (10.4)$$

Thus exactly half of the odd positions in every nonzero Pascal row have even binary weight and half have odd binary weight.

Proof. Set $z = 1$ in (10.2). There are $2^{\text{wt}_2(n)}$ support positions by theorem 5.2. $\square$

Theorem 10.3 (Sparse Prouhet theorem). *Let $n > 0$, $s = \text{wt}_2(n)$. For every polynomial p of degree below s ,*

$$\sum_{k \preceq_2 n} \varepsilon_k p(k) = 0. \quad (10.5)$$

The first surviving power moment is

$$\sum_{k \preceq_2 n} \varepsilon_k k^s = (-1)^s s! 2^{\beta(n)}. \quad (10.6)$$

Proof. Every factor in (10.2) has a simple zero at $z = 1$, so R_n has a zero of exact order s , with local leading term

$$R_n(z) = (-1)^s 2^{\beta(n)} (z - 1)^s + \mathcal{O}((z - 1)^{s+1}).$$

The derivative/falling-factorial proof of [theorem 9.5](#) applies with the sum restricted to submasks. $\square$

Formal versions. In `ThueMorseSparseProuhet.lean` the submasks of n are parametrized by the subsets of the bit support `Fabius.bitSupport` (with dictionary `Fabius.mem_bitSupport`, `Fabius.sum_two_pow_bitSupport`, `Fabius.card_bitSupport`); the sparse cancellation and the sharp moment are `Fabius.sum_submask_neg_one_pow_eval` and `Fabius.sum_submask_neg_one_pow_pow`, both instances of the arbitrary-weight engine of [theorem 9.4](#) with $w_j = 2^j h$ running over $J(n)$, over any commutative ring.

Corollary 10.4 (Sparse operator identity). *For every function f ,*

$$\sum_{k \preceq_2 n} \varepsilon_k f(x + hk) = \prod_{j \in J(n)} (I - T_{2^j h}) f(x). \quad (10.7)$$

If $f \in C^s$ and $h \geq 0$, its absolute value is at most

$$2^{\beta(n)} h^s \sup_{x \leq y \leq x + nh} |f^{(s)}(y)|. \quad (10.8)$$

Proof. Expand the product over the set-bit positions and apply the fundamental theorem of calculus once for each step. $\square$

The whole sparse moment sequence also has a direct finite formula.

Corollary 10.5 (All sparse moments). *For $r \geq s$,*

$$\sum_{k \preceq_2 n} \varepsilon_k k^r = (-1)^s r! 2^{\beta(n)} \sum_{\substack{(q_j)_{j \in J(n)} \geq 0 \\ \sum_{j \in J(n)} q_j = r - s}} \prod_{j \in J(n)} \frac{2^{jq_j}}{(q_j + 1)!}. \quad (10.9)$$

Proof. Substitute $z = e^t$ in (10.2), factor $-2^j t$ from every occupied bit, and extract coefficients exactly as in [theorem 9.7](#). $\square$

Example 10.6. For $n = 13 = (1101)_2$, $J(13) = \{0, 2, 3\}$, $s = 3$, and $\beta(13) = 5$. The odd positions in row 13 are

$$0, 1, 4, 5, 8, 9, 12, 13.$$

Their signed sums against $1, k, k^2$ vanish, while

$$\sum_{k \preceq_2 13} \varepsilon_k k^3 = (-1)^3 3! 2^5 = -192.$$

This is a Prouhet partition on a sparse, Pascal-determined grid rather than a consecutive interval.

11 Exact prefix sums, iterated antiderivatives, and the Fabius bridge

11.1 The first prefix discrepancy

Define the exclusive signed prefix and the number of one-bits in the sequence prefix by

$$S(N) = \sum_{n=0}^{N-1} \varepsilon_n, \quad A(N) = \sum_{n=0}^{N-1} \tau(n). \quad (11.1)$$

Theorem 11.1 (Exact prefix formula). *For every $q \geq 0$,*

$$\boxed{S(2q) = 0, \quad S(2q+1) = \varepsilon_q.} \quad (11.2)$$

Equivalently,

$$S(N) = \frac{1 - (-1)^N}{2} \varepsilon_{\lfloor N/2 \rfloor}. \quad (11.3)$$

Therefore

$$\boxed{A(N) = \frac{N - S(N)}{2}} \quad (11.4)$$

and explicitly

$$A(2q) = q, \quad A(2q+1) = q + \tau(q). \quad (11.5)$$

Proof. Every adjacent pair cancels: $\varepsilon_{2j} + \varepsilon_{2j+1} = \varepsilon_j - \varepsilon_j = 0$. An odd-length prefix leaves the unpaired term $\varepsilon_{2q} = \varepsilon_q$. The count formula follows from $\varepsilon_n = 1 - 2\tau(n)$. $\square$

The inclusive sum through N is

$$\sum_{n=0}^N \varepsilon_n = \frac{1 + (-1)^N}{2} \varepsilon_{\lfloor N/2 \rfloor}. \quad (11.6)$$

Corollary 11.2 (Interval discrepancy). *For $0 \leq a \leq b$,*

$$\sum_{n=a}^{b-1} \varepsilon_n = S(b) - S(a), \quad (11.7)$$

$$\left| \sum_{n=a}^{b-1} \varepsilon_n \right| \leq 2, \quad (11.8)$$

$$\left| \sum_{n=a}^{b-1} \tau(n) - \frac{b-a}{2} \right| \leq 1. \quad (11.9)$$

Both constants are attained.

Proof. The interval sum is a difference of two values in $\{-1, 0, 1\}$. Convert to zero–one values by (1.3). For sharpness, $\varepsilon_5 + \varepsilon_6 = 2$. $\square$

Corollary 11.3 (Aligned and partial dyadic blocks). *For $m \geq 1$ and $q \geq 0$,*

$$\sum_{r=0}^{2^m-1} \varepsilon_{2^m q+r} = 0. \quad (11.10)$$

More generally, for $0 \leq R \leq 2^m$,

$$\sum_{r=0}^{R-1} \varepsilon_{2^m q+r} = \varepsilon_q S(R). \quad (11.11)$$

Proof. Use (2.6) and factor out ε_q ; then apply theorem 11.1. $\square$

The prefix generating function is unusually simple:

$$\sum_{N \geq 0} S(N)z^N = \frac{z}{1-z} E(z) = zE(z^2) = z \prod_{j \geq 1} (1 - z^{2^j}). \quad (11.12)$$

The middle equality is the Mahler equation.

11.2 Iterated inclusive prefixes

Define discrete antiderivatives by

$$S_n^{(0)} = \varepsilon_n, \quad S_n^{(k+1)} = \sum_{j=0}^n S_j^{(k)}. \quad (11.13)$$

Then (11.6) says

$$S_{2q}^{(1)} = \varepsilon_q, \quad S_{2q+1}^{(1)} = 0. \quad (11.14)$$

Theorem 11.4 (Binomial convolution for discrete antiderivatives). *For $k \geq 1$,*

$$S_n^{(k)} = \sum_{j=0}^n \binom{n-j+k-1}{k-1} \varepsilon_j. \quad (11.15)$$

Its ordinary generating function is

$$\sum_{n \geq 0} S_n^{(k)} z^n = \frac{E(z)}{(1-z)^k}. \quad (11.16)$$

Proof. One inclusive summation multiplies an ordinary generating function by $(1-z)^{-1}$. Iterate k times, then expand $(1-z)^{-k} = \sum_{r \geq 0} \binom{r+k-1}{k-1} z^r$ and take the Cauchy product. $\square$

11.3 Terminal zero runs at dyadic boundaries

The quotient of a finite Prouhet polynomial by $(1-z)^k$ is still a polynomial when $k \leq r$:

$$\frac{P_r(z)}{(1-z)^k} = (1-z)^{r-k} \prod_{j=0}^{r-1} (1+z+\cdots+z^{2^j-1}). \quad (11.17)$$

Its degree is $2^r - k - 1$.

Theorem 11.5 (Dyadic terminal zero run). *If $1 \leq k \leq r$, then*

$$S_{2^r-k+i}^{(k)} = 0, \quad 0 \leq i < k. \quad (11.18)$$

The adjacent values are

$$S_{2^r-k-1}^{(k)} = (-1)^{r-k}, \quad S_{2^r}^{(k)} = -1. \quad (11.19)$$

Proof. Below degree 2^r , the coefficients of $E(z)/(1-z)^k$ agree with those of $P_r(z)/(1-z)^k$. The latter has degree $2^r - k - 1$, proving the zero run. Its leading coefficient is $(-1)^{r-k}$, giving the preceding value. The coefficient at 2^r comes from the first omitted factor $1 - z^{2^r}$, or directly from the prefix recurrence. $\square$

11.4 Why this is the discrete spine of the Fabius theory

The chain connecting Thue–Morse signs to the Fabius and Rvachev functions is now algebraically visible:

$$\boxed{\begin{aligned} \text{binary parity character} &\implies P_r(z) = \prod_{j < r} (1 - z^{2^j}) \\ &\implies \frac{P_r(z)}{(1 - z)^k} \implies \text{iterated-prefix coefficient grids} \end{aligned}} \quad (11.20)$$

The zero of order k at $z = 1$ creates the exact terminal flatness in [theorem 11.5](#); after the normalizations developed in the main repository exposition, the resulting coefficient grids and polygonal interpolants converge to the Fabius function. On the Fourier side, replacing z by a point on the unit circle turns the same finite product into a finite product of sine factors, whose scaled limit is the sinc product associated with the Rvachev up-function.

Thus the discrete-prefix and analytic-sinc appearances of Thue–Morse are not separate phenomena. They are the coefficient and Fourier-evaluation faces of the single factorization (6.3). The quantitative convergence theory belongs to the dedicated companion reports and Lean modules; the purpose here is to expose the exact common algebraic spine.

12 Ordinary Fourier transforms and product-to-sum identities

12.1 A finite Fourier product

Putting $z = e^{i\theta}$ in (6.3) gives

$$\sum_{n=0}^{2^m-1} \varepsilon_n e^{in\theta} = \prod_{j=0}^{m-1} (1 - e^{i2^j\theta}). \quad (12.1)$$

Using $1 - e^{iu} = -2ie^{iu/2} \sin(u/2)$,

$$\boxed{\sum_{n=0}^{2^m-1} \varepsilon_n e^{in\theta} = (-2i)^m e^{i(2^m-1)\theta/2} \prod_{j=0}^{m-1} \sin(2^{j-1}\theta).} \quad (12.2)$$

Hence

$$\left| \sum_{n=0}^{2^m-1} \varepsilon_n e^{in\theta} \right| = 2^m \prod_{j=0}^{m-1} |\sin(2^{j-1}\theta)|. \quad (12.3)$$

12.2 Sine and cosine products as Thue–Morse sums

A rescaling of (12.2) gives exact finite product-to-sum identities.

Theorem 12.1 (Thue–Morse trigonometric sums). *For every integer $m \geq 0$ and every complex z ,*

$$\boxed{\prod_{r=0}^m \sin \frac{z}{2^r} = \frac{1}{2^m} \sum_{n=0}^{2^m-1} \varepsilon_n \sin \left(\frac{\pi n}{2} + \frac{(2n+1)z}{2^m} \right).} \quad (12.4)$$

Also

$$\boxed{\cos z \prod_{r=1}^m \sin \frac{z}{2^r} = \frac{1}{2^m} \sum_{n=0}^{2^m-1} \varepsilon_n \cos \left(\frac{\pi n}{2} + \frac{(2n+1)z}{2^m} \right).} \quad (12.5)$$

The product from $r = 1$ to 0 is empty when $m = 0$.

Proof. Set $\theta = 2z/2^m$ in (12.2), multiply by $i^m e^{iz/2^m}$, and simplify the phase. The result is

$$\sum_{n < 2^m} \varepsilon_n \exp\left(i \left(\frac{\pi m}{2} + \frac{(2n+1)z}{2^m} \right)\right) = 2^m e^{iz} \prod_{r=1}^m \sin \frac{z}{2^r}.$$

For real z , take imaginary and real parts. Both resulting identities are identities between entire functions of z , so the identity theorem extends them to every complex z . $\square$

These identities make explicit the Fourier route from finite Thue–Morse blocks to the sinc products in the Fabius/Rvachev analysis.

12.3 The exact dyadic discrete Fourier transform

Let

$$\widehat{\varepsilon}_m(k) = \sum_{n=0}^{2^m-1} \varepsilon_n \exp\left(-\frac{2\pi i k n}{2^m}\right), \quad 0 \leq k < 2^m. \quad (12.6)$$

Theorem 12.2 (Dyadic DFT). *For every k ,*

$$\widehat{\varepsilon}_m(k) = \prod_{j=0}^{m-1} \left(1 - \exp\left(-\frac{2\pi i k 2^j}{2^m}\right) \right). \quad (12.7)$$

If $m \geq 1$, this is zero for every even k . For odd k ,

$$\widehat{\varepsilon}_m(k) = (2i)^m e^{-\pi i k (1-2^{-m})} \prod_{r=1}^m \sin \frac{\pi k}{2^r}. \quad (12.8)$$

Thus

$$|\widehat{\varepsilon}_m(k)| = 2^m \prod_{r=1}^m \left| \sin \frac{\pi k}{2^r} \right| \quad (k \text{ odd}). \quad (12.9)$$

Proof. Evaluate (6.3) at $z = e^{-2\pi i k / 2^m}$. If k is even, the factor with $j = m-1$ is $1 - (-1)^k = 0$. For odd k , apply $1 - e^{-iu} = 2ie^{-iu/2} \sin(u/2)$ factorwise and sum the phases. $\square$

For $m \geq 1$, Fourier inversion gives a root-of-unity formula for every individual term:

$$\varepsilon_n = \frac{1}{2^m} \sum_{\substack{0 \leq k < 2^m \\ k \text{ odd}}} \widehat{\varepsilon}_m(k) e^{2\pi i k n / 2^m}, \quad 0 \leq n < 2^m. \quad (12.10)$$

This is intentionally extravagant as an evaluator, but exact and natural for spectral calculations.

13 Walsh characters, Rademacher functions, Riesz products, and autocorrelation

13.1 A one-point Walsh spectrum

For m -bit vectors $a, b \in \mathbb{F}_2^m$, write $\langle a, b \rangle = \sum_j a_j b_j \pmod{2}$, and let $\mathbf{1} = (1, \dots, 1)$. For $n < 2^m$,

$$\varepsilon_n = (-1)^{\langle \mathbf{1}, \mathbf{b}(n) \rangle}. \quad (13.1)$$

Theorem 13.1 (Walsh transform is a point mass). *For every $a \in \mathbb{F}_2^m$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n (-1)^{\langle a, b(n) \rangle} = \begin{cases} 2^m, & a = \mathbf{1}, \\ 0, & a \neq \mathbf{1}. \end{cases} \quad (13.2)$$

Proof. The summand is the character indexed by $a + \mathbf{1}$. A nontrivial character sums to zero over the finite Boolean group, while the trivial character sums to its order 2^m . $\square$

Thus the broad ordinary Fourier spectrum is a basis-change phenomenon: the same block is spectrally pure in the Walsh basis.

13.2 Rademacher step functions

For $x \in [0, 1)$, define the right-continuous Rademacher functions

$$r_j(x) = (-1)^{\lfloor 2^{j+1}x \rfloor}. \quad (13.3)$$

If $0 \leq n < 2^m$ and $x \in [n/2^m, (n+1)/2^m)$, then the first m binary digits of x encode n , so

$$\varepsilon_n = \prod_{j=0}^{m-1} r_j(x). \quad (13.4)$$

The finite Thue–Morse block is therefore the sampling of one Walsh function on the dyadic partition.

13.3 Finite Riesz products

Define

$$\rho_m(x) = 2^{-m} |P_m(e^{2\pi i x})|^2. \quad (13.5)$$

Theorem 13.2 (Riesz-product density). *For every $m \geq 0$,*

$$\rho_m(x) = \prod_{j=0}^{m-1} (1 - \cos(2\pi 2^j x)). \quad (13.6)$$

Moreover,

$$\int_0^1 \rho_m(x) dx = 1, \quad (13.7)$$

so $\rho_m(x) dx$ is a probability measure.

Proof. Use $|1 - e^{i\theta}|^2 = 2(1 - \cos \theta)$ in (6.3). Parseval gives

$$\int_0^1 \rho_m(x) dx = 2^{-m} \sum_{n < 2^m} |\varepsilon_n|^2 = 1.$$

$\square$

The finite measures converge weakly to the Thue–Morse spectral measure. The formal notation

$$d\mu_{\text{TM}}(x) = \prod_{j \geq 0} (1 - \cos(2\pi 2^j x)) dx \quad (13.8)$$

means this weak limit, not pointwise multiplication of ordinary densities. The limiting measure is purely singular continuous, a deeper theorem of substitution spectral theory [14, 15].

13.4 Finite and limiting autocorrelation

For $m, k \geq 0$, define the dyadic average

$$\eta_m(k) = \frac{1}{2^m} \sum_{n=0}^{2^m-1} \varepsilon_n \varepsilon_{n+k}, \quad (13.9)$$

where the second index is not reduced modulo 2^m .

Proposition 13.3 (Finite autocorrelation recursion). *For $r \geq 0$,*

$$\eta_{m+1}(2r) = \eta_m(r), \quad (13.10)$$

$$\eta_{m+1}(2r+1) = -\frac{\eta_m(r) + \eta_m(r+1)}{2}. \quad (13.11)$$

Proof. Split the defining sum into even and odd n . For an even shift, both halves reduce to $\varepsilon_j \varepsilon_{j+r}$. For an odd shift, they reduce to $-\varepsilon_j \varepsilon_{j+r}$ and $-\varepsilon_j \varepsilon_{j+r+1}$, respectively. $\square$

Theorem 13.4 (Limiting autocorrelation). *For every fixed $k \geq 0$, the limit*

$$\eta(k) = \lim_{m \rightarrow \infty} \eta_m(k) \quad (13.12)$$

exists and is characterized by

$$\eta(0) = 1, \quad (13.13)$$

$$\eta(2r) = \eta(r), \quad (13.14)$$

$$\eta(2r+1) = -\frac{\eta(r) + \eta(r+1)}{2}. \quad (13.15)$$

In particular,

$$\eta(1) = -\frac{1}{3}, \quad \eta(2) = -\frac{1}{3}, \quad \eta(3) = \frac{1}{3}, \quad \eta(5) = 0. \quad (13.16)$$

Proof. Induct on k . The case $k = 0$ is constant. For $k = 1$, (13.11) gives the contraction

$$\eta_{m+1}(1) = -\frac{1 + \eta_m(1)}{2},$$

whose fixed point is $-1/3$. If $k = 2r > 1$, convergence reduces to the smaller index r . If $k = 2r+1 > 1$, both r and $r+1$ are smaller than k , so the odd recurrence and the induction hypothesis give convergence. Pass to the limit and evaluate recursively. $\square$

The numbers $\eta(k)$ are the Fourier coefficients of μ_{TM} . Thus the substitution recursion, autocorrelation, and Riesz product are exact combinatorial, coefficient, and measure-theoretic forms of the same spectral object. Extending by evenness, $\eta(-k) = \eta(k)$, the first values are

$$1, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}, 0, \frac{1}{3}, 0, -\frac{1}{3}, \frac{1}{6}, 0, -\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}, 0, \frac{1}{6}. \quad (13.17)$$

13.5 A Mahler equation for the autocorrelation series

The autocorrelation sequence has its own generating function, and the recursion of [theorem 13.4](#) compresses into a single functional equation of Mahler type.

Theorem 13.5 (Autocorrelation Mahler equation). *Let $C(z) = \sum_{n \geq 0} \eta(n) z^n$ for $|z| < 1$. Then*

$$\boxed{2zC(z) = 1 - (1-z)^2 C(z^2)}. \quad (13.18)$$

Proof. Split the series by parity of the index and insert (13.14) and (13.15):

$$\begin{aligned}
 C(z) &= \sum_{n \geq 0} \eta(2n)z^{2n} + \sum_{n \geq 0} \eta(2n+1)z^{2n+1} \\
 &= C(z^2) - \frac{z}{2} \sum_{n \geq 0} \eta(n)z^{2n} - \frac{z}{2} \sum_{n \geq 0} \eta(n+1)z^{2n} \\
 &= C(z^2) - \frac{z}{2} C(z^2) - \frac{1}{2z} (C(z^2) - 1).
 \end{aligned}$$

Multiply by $2z$ and collect the $C(z^2)$ terms into $-(1-z)^2 C(z^2)$. $\square$

Equation (13.18) places the autocorrelation beside E and T in the same Mahler framework; the square $(1-z)^2$ reflects the fact that η averages products of two signs. Since the $\eta(k)$ are also the Fourier coefficients of the spectral measure, the equation is a coefficient-level renormalization law for that singular continuous measure: the measure has no closed-form distribution function, yet its Fourier data satisfies a two-term finite recursion.

14 Combinatorics and dynamics of the infinite word

The sequence read as an infinite binary word $\mathbf{t} = \tau(0)\tau(1)\tau(2)\cdots$ has a second life in combinatorics on words. A *factor* of $\mathbf{t}$ is a finite block $\tau(i)\tau(i+1)\cdots\tau(i+\ell-1)$ occurring at some position i ; the word-combinatorial theory studies which factors occur, how often, and with which repetitions. This section proves the elementary statements in full and states the two classical deep ones with references.

14.1 Uniform recurrence and letter frequencies

Recall from (2.4) that $\mathbf{t}$ is the fixed point, starting with 0, of the substitution $\mu : 0 \mapsto 01, 1 \mapsto 10$. Iterating, $\mathbf{t} = \lim_m \mu^m(0)$, and $\mu^m(0), \mu^m(1)$ are the length- 2^m blocks $W_m, \overline{W}_m$ of (2.3), called the level- m supertiles.

Proposition 14.1 (Uniform recurrence). *The word $\mathbf{t}$ is uniformly recurrent: for every factor w there is a length $L(w)$ such that every window of $L(w)$ consecutive letters of $\mathbf{t}$ contains an occurrence of w .*

Proof. A factor w of length ℓ occurs inside some supertile $\mu^k(a)$ with $2^k \geq \ell$ and $a \in \{0, 1\}$. Both letters occur in $\mu(0) = 01$, so every supertile of level $k+1$ contains both $\mu^k(0)$ and $\mu^k(1)$; in particular every level- $(k+1)$ supertile contains w . Since $\mathbf{t}$ is tiled by consecutive level- $(k+1)$ supertiles, every window of length $L(w) = 2^{k+2}$ contains a complete one. $\square$

Uniform recurrence says every pattern that occurs once occurs with bounded gaps. In dynamical terms, the closure of the shift orbit of $\mathbf{t}$ is a minimal, uniquely ergodic system; that standard consequence of primitive constant-length substitutions [14] is what guarantees the existence of the limiting frequencies and autocorrelations used above. The two letters have frequency exactly $\frac{1}{2}$: at dyadic lengths this is the exact balance $A(2q) = q$ of (11.5), and general lengths deviate by at most one by theorem 11.2.

14.2 Aperiodicity

Proposition 14.2 (No eventual period). *The sequence (ε_n) is not eventually periodic. In particular it is not periodic.*

Proof. Suppose $p \geq 1$ were a period valid from some index N_0 on. For any $r \geq 0$, choose m with $2^m > r + p$ and $2^m + r \geq N_0$. Block concatenation (2.6) gives $\varepsilon_{2^m+r} = -\varepsilon_r$ and $\varepsilon_{2^m+r+p} = -\varepsilon_{r+p}$; eventual periodicity equates the left sides, so $\varepsilon_{r+p} = \varepsilon_r$ for every $r \geq 0$, and p is a global period. Now compare reflections: by (2.8), for every m with $2^m > p$,

$$-1 = \varepsilon_{2^m} = \varepsilon_{2^m-p} = \varepsilon_{(2^m-1)-(p-1)} = (-1)^m \varepsilon_{p-1}.$$

The right side changes sign between consecutive m , while the left side is constant — a contradiction. $\square$

Formal versions. `ThueMorseAperiodicity.lean` follows this argument verbatim: `Fabius.t hueMorseSign_two_pow` and the block-flip `Fabius.thueMorseSign_two_pow_add` transport an eventual period back to a global one (`Fabius.thueMorseSign_period_of_eventually`), and `Fabius.thueMorseSign_not_eventually_periodic` plays the reflection at two consecutive block levels exactly as above.

14.3 Overlap-freeness and the critical exponent

A *square* is a nonempty word repeated twice in immediate succession, ww ; a *cube* is www ; an *overlap* is a word of the form $axaxa$ with a a single letter and x a possibly empty word — equivalently, a square ww prolonged by one further letter of the same period. The *critical exponent* of an infinite word is the supremum, over its factors w , of $|w|/p$, where p is the smallest period of w and $|w|$ its length.

Theorem 14.3 (Thue; literature theorem). *The Thue–Morse word contains no overlap. Consequently it contains no cube, and no factor has exponent strictly greater than 2; since squares of arbitrarily great length do occur (apply μ^m to the square $\tau(1)\tau(2) = 11$), the critical exponent equals*

$$\boxed{2}, \tag{14.1}$$

and it is attained.

This is Thue’s foundational theorem [3]; a modern proof by desubstitution — a shortest overlap in $\mu(\mathbf{t})$ would force a shorter one in $\mathbf{t}$ after a finite boundary check — appears in [12]. Overlap-freeness is far stronger than the directly visible absence of the cubes 000 and 111, and it was the original reason for the word’s construction, decades before its automatic, spectral, and Diophantine roles emerged.

14.4 Complement and reversal closure

Complementing every letter commutes with the substitution up to swapping the seeds $0 \leftrightarrow 1$; hence with every factor w the language of $\mathbf{t}$ also contains its bitwise complement $\bar{w}$. For reversal, the reflection formula (2.8) says that reading the length- 2^m prefix backwards reproduces it when m is even and complements it when m is odd — in signed terms this is the reciprocity of theorem 6.4. Every factor sits inside some such prefix, so the factor language is closed under reversal as well. By unique ergodicity, complementary factors occur with equal frequencies; the exact prefix balance of theorem 11.1 is the one-letter case.

14.5 Linear factor complexity

Let $p(\ell)$ denote the *factor complexity*: the number of distinct factors of length ℓ .

Proposition 14.4 (Elementary linear bound). *For every $\ell \geq 2$,*

$$\boxed{\ell + 1 \leq p(\ell) < 8\ell.} \tag{14.2}$$

Proof. Choose k minimal with $2^k \geq \ell$, so $2^k < 2\ell$. The word is a concatenation of level- k supertiles, so every length- ℓ factor lies inside a pair of consecutive supertiles $\mu^k(a)\mu^k(b)$. There are at most four choices of (a, b) and at most 2^k starting offsets, whence $p(\ell) \leq 4 \cdot 2^k < 8\ell$. The lower bound is the Morse–Hedlund theorem [19]: an infinite word with $p(\ell) \leq \ell$ for some ℓ is eventually periodic, which [theorem 14.2](#) forbids; hence $p(\ell) \geq \ell + 1$ for all ℓ . $\square$

Linear complexity has two immediate consequences. The topological entropy, $\lim_{\ell} \ell^{-1} \log p(\ell)$, is zero. And the word is not *normal*: a normal binary word contains all 2^ℓ blocks of every length ℓ , which is incompatible with (14.2). Exact piecewise-linear formulas for $p(\ell)$ are known [12]; the elementary bound above suffices for every use in this atlas. The pointed contrast with [section 15](#) below is worth savoring: the word itself is combinatorially rigid, yet its arithmetic subsequences behave pseudorandomly.

15 Rarefied sums, arithmetic progressions, and Newman’s phenomenon

Prefix sums of ε are bounded by [theorem 11.1](#), and sums over aligned dyadic blocks vanish. Restricting instead to an arithmetic progression destroys this balance in a striking way: along multiples of three one sign is *always* in excess, with a power-law surplus. The finite product P_m evaluated at roots of unity controls the whole phenomenon.

15.1 Root-of-unity filters at dyadic endpoints

For a modulus $q \geq 2$ and residue $0 \leq r < q$, define

$$A_{m,r}^{(q)} = \sum_{\substack{0 \leq n < 2^m \\ n \equiv r \pmod{q}}} \varepsilon_n. \quad (15.1)$$

Let $\zeta_q = e^{2\pi i/q}$, a primitive q -th root of unity. Averaging the geometric identity $\frac{1}{q} \sum_{\ell < q} \zeta_q^{(n-r)\ell} = \mathbf{1}_{q|n-r}$ against (6.3) gives the exact finite formula

$$A_{m,r}^{(q)} = \frac{1}{q} \sum_{\ell=0}^{q-1} \zeta_q^{-r\ell} P_m(\zeta_q^\ell) = \frac{1}{q} \sum_{\ell=0}^{q-1} \zeta_q^{-r\ell} \prod_{j=0}^{m-1} (1 - \zeta_q^{\ell 2^j}). \quad (15.2)$$

For $m \geq 1$ the term $\ell = 0$ vanishes, and the exponents $\ell 2^j$ move along the orbit of multiplication by two modulo q . For *even* moduli the phenomenon collapses: powers of two are eventually $0 \pmod{2^k}$, and for example $A_{m,0}^{(2)} = \sum_{k < 2^{m-1}} \varepsilon_{2k} = S(2^{m-1}) = 0$ for $m \geq 2$ by (11.2). Odd moduli are the interesting case: there the orbit of 2 is a fixed cycle and the products in (15.2) grow geometrically.

15.2 Complete evaluation modulo three

For $q = 3$ everything is explicit, because $2^j \equiv 1, 2, 1, 2, \dots \pmod{3}$ and, with $\omega = \zeta_3 = e^{2\pi i/3}$,

$$(1 - \omega)(1 - \omega^2) = 3. \quad (15.3)$$

Theorem 15.1 (Exact dyadic rarefaction modulo three). *For $u \geq 1$,*

$$(A_{2u,0}^{(3)}, A_{2u,1}^{(3)}, A_{2u,2}^{(3)}) = 3^{u-1} (2, -1, -1), \quad (15.4)$$

and for $u \geq 0$,

$$(A_{2u+1,0}^{(3)}, A_{2u+1,1}^{(3)}, A_{2u+1,2}^{(3)}) = 3^u (1, -1, 0). \quad (15.5)$$

Proof. At even level $m = 2u$, the factors of $P_m(\omega)$ pair into u copies of (15.3), so $P_{2u}(\omega) = P_{2u}(\omega^2) = 3^u$. At odd level one factor is left over: $P_{2u+1}(\omega) = 3^u(1 - \omega)$ and $P_{2u+1}(\omega^2) = 3^u(1 - \omega^2)$. Insert these values into (15.2) with $q = 3$ and simplify with $1 + \omega + \omega^2 = 0$. $\square$

Formal versions. `ThueMorseRarefied.lean` proves the closed vectors (15.4)–(15.5) as `Fabius.rarefiedSum_three_closed` – and does so without any root of unity: the one-step halving recursion $A_{m+1,r} = A_{m,2r \bmod 3} - A_{m,(2r+1) \bmod 3}$ (`Fabius.rarefiedSum_three_succ`; the hypothesis $r < 3$ is necessary, as the recursion is false for out-of-range residues) is iterated over paired levels by pure integer induction, so the filter argument above is subsumed rather than formalized. Newman positivity at dyadic endpoints is `Fabius.rarefiedSum_three_zero_pos`, the full-block balance is `Fabius.sum_thueMorseSign_range_two_pow`, and the even-modulus collapse of the surrounding text is `Fabius.rarefiedSum_two_zero`.

The first vectors are

$$(1, -1, 0), \quad (2, -1, -1), \quad (3, -3, 0), \quad (6, -3, -3), \quad (9, -9, 0),$$

growing by a factor of three every two levels. Writing $X = 2^m$ for the endpoint, the excess on multiples of three is

$$A_{m,0}^{(3)} = c_m X^\alpha, \quad \alpha = \log_4 3 = \frac{\log 3}{\log 4} = 0.79248\dots, \quad (15.6)$$

where the constant c_m equals $\frac{2}{3}$ for even m and $\frac{1}{\sqrt{3}}$ for odd m : a clean two-scale oscillation around the power law.

15.3 Gelfond’s uniform exponent

The rarefied sums are finitely many samples of the trigonometric polynomial $p_m(x) = P_m(e^{2\pi i x})$, whose modulus is the sine product (12.3). How large can it get? Evaluating at $x = \frac{1}{3}$ gives, by (15.3),

$$\sup_{x \in \mathbb{R}} |p_m(x)| \geq |P_m(\omega)| = 3^{m/2} \quad (m \geq 0), \quad (15.7)$$

since the factors have modulus $|1 - \omega| = |1 - \omega^2| = \sqrt{3}$ in every position. Gelfond’s theorem [28] shows this is sharp up to a bounded factor:

$$\sup_{x \in \mathbb{R}} |p_m(x)| \leq \frac{2}{\sqrt{3}} 3^{m/2} = \frac{2}{\sqrt{3}} (2^m)^{\log_4 3}. \quad (15.8)$$

Decomposing an arbitrary range $[0, N)$ into at most one dyadic block per binary digit of N and applying (15.8) blockwise gives

$$\sup_{x \in \mathbb{R}} \left| \sum_{n=0}^{N-1} \varepsilon_n e^{2\pi i n x} \right| = \mathcal{O}(N^{\log_4 3}), \quad (15.9)$$

so $\log_4 3$ is the optimal uniform growth exponent of Thue–Morse exponential sums [28, 29]. At $x = 0$ the sum collapses to the bounded prefix values – the supremum and the special value differ by a full power of N .

15.4 A matrix recursion for odd moduli

For a fixed odd q , the vector $\mathbf{A}_m = (A_{m,r}^{(q)})_{0 \leq r < q}$ obeys a linear recursion. Splitting (15.1) by the parity of n , and writing 2^{-1} for the inverse of 2 modulo q (which exists because q is odd),

$$A_{m+1,r}^{(q)} = A_{m,2^{-1}r \bmod q}^{(q)} - A_{m,2^{-1}(r-1) \bmod q}^{(q)}, \quad (15.10)$$

since even $n = 2n' \equiv r$ forces $n' \equiv 2^{-1}r$ with $\varepsilon_n = \varepsilon_{n'}$, and odd $n = 2n' + 1 \equiv r$ forces $n' \equiv 2^{-1}(r - 1)$ with $\varepsilon_n = -\varepsilon_{n'}$. Thus $\mathbf{A}_{m+1} = M_q \mathbf{A}_m$ for a fixed matrix M_q with one entry $+1$ and one entry -1 per row; diagonalizing M_q by the discrete Fourier transform on $\mathbb{Z}/q\mathbb{Z}$ reproduces (15.2). For $q = 3$ the eigenvalues have modulus $\sqrt{3}$, which is the growth rate seen in theorem 15.1.

15.5 The global Newman theorem and the arithmetic frontier

Theorem 15.1 evaluates the excess at dyadic endpoints. Newman’s theorem extends the positivity to *every* endpoint. The following results are literature theorems, stated here to locate the atlas’s exact identities inside modern analytic number theory; none is reproved.

Theorem 15.2 (Newman; literature theorem). *For every real $X \geq 1$,*

$$N_3(X) := \sum_{\substack{0 \leq n < X \\ 3|n}} \varepsilon_n > 0. \quad (15.11)$$

Moreover $\frac{1}{20} X^{\log_4 3} \leq N_3(X) \leq 5X^{\log_4 3}$ for $X \geq 2$.

Among multiples of three below any cutoff, integers of even binary weight always outnumber those of odd binary weight [30]. Coquet determined the precise structure: $N_3(X) = X^{\log_4 3} \Psi(\log_4 X) + \mathcal{O}(1)$ with Ψ a continuous, nowhere-differentiable periodic function [31]. Rarefactions along other progressions and for generalized digit characters were analyzed by Drmota and Skalba [32] and Drmota and Stoll [33]; a recent automata-theoretic treatment mechanizes such evaluations [34].

Beyond arithmetic progressions, the same digit-parity character is now understood along much thinner sequences. Mauduit and Rivat proved Gelfond’s conjecture that the parity is equidistributed along the primes: $\sum_{p \leq x} \varepsilon_p = o(\pi(x))$, the sum running over primes with $\pi(x)$ their count [35]. Drmota, Mauduit, and Rivat proved that along the squares the word $(\tau(n^2))_{n \geq 0}$ is *normal* – every binary block of length L occurs with limiting frequency 2^{-L} [36] – in sharp contrast with the non-normality of the word itself (theorem 14.4). Spiegelhofer established a level-of-distribution result covering $\tau(\lfloor n^c \rfloor)$ for $1 < c < 2$ [37] and the one-symbol balance along the cubes [38]. The deterministic word is combinatorially rigid, yet its polynomial and prime subsequences behave like fair coin flips: that tension is the modern face of Gelfond’s problems, and every one of these theorems begins from the finite product and carry structure catalogued in this atlas.

16 Boundary flatness, shifted Dirichlet series, and Mellin continuation

This section develops an analytic consequence not organized systematically in the four drafts. The lacunary product is extraordinarily flat as its argument approaches one from inside the unit disk. That flatness removes every Mellin singularity and gives an entire continuation of a natural family of Thue–Morse Dirichlet series.

16.1 Quadratic logarithmic decay near one

For $t > 0$, set

$$\mathcal{E}(t) = E(e^{-t}) = \prod_{j \geq 0} (1 - e^{-2^j t}). \quad (16.1)$$

Every factor lies in $(0, 1)$, so $\mathcal{E}(t) > 0$.

Theorem 16.1 (Boundary flatness). *As $t \downarrow 0$,*

$$\log \mathcal{E}(t) = -\frac{\log^2(1/t)}{2 \log 2} + \mathcal{O}(\log(1/t)). \quad (16.2)$$

Consequently, for every $A > 0$,

$$\mathcal{E}(t) = \mathcal{O}_A(t^A) \quad (t \downarrow 0). \quad (16.3)$$

Proof. Assume $0 < t < 1/2$, put $L = \log(1/t)$, and let $m = \lfloor \log_2(1/t) \rfloor$. Then $2^m t \in (1/2, 1]$. For $0 \leq j < m$, one has $0 < 2^j t \leq 1/2$, and the elementary bounds

$$\frac{x}{2} \leq 1 - e^{-x} \leq x \quad (0 < x \leq 1) \quad (16.4)$$

show that

$$\log(1 - e^{-2^j t}) = \log t + j \log 2 + \mathcal{O}(1), \quad (16.5)$$

with an absolute $\mathcal{O}(1)$. Summing gives

$$\sum_{j=0}^{m-1} \log(1 - e^{-2^j t}) = m \log t + \frac{m(m-1)}{2} \log 2 + \mathcal{O}(m). \quad (16.6)$$

For the tail, write $2^m t = u \in (1/2, 1]$. Then

$$\sum_{j=m}^{\infty} \left| \log(1 - e^{-2^j t}) \right| = \sum_{r=0}^{\infty} \left| \log(1 - e^{-2^r u}) \right| = \mathcal{O}(1) \quad (16.7)$$

uniformly in u , because the terms decay doubly exponentially. Finally $m = L/\log 2 + \mathcal{O}(1)$; substitution in (16.6) yields (16.2). The negative quadratic term dominates $-A \log(1/t)$ for every fixed A , proving (16.3). $\square$

Remark 16.2. Sharper asymptotic expansions contain a bounded log-periodic fluctuation in $\log_2(1/t)$. The elementary estimate above is deliberately limited to the part needed for Mellin continuation: super-polynomial flatness.

16.2 The shifted Dirichlet family

For $a > 0$, define initially in the half-plane $\Re s > 0$

$$D(s, a) = \sum_{n=0}^{\infty} \frac{\varepsilon_n}{(n+a)^s}. \quad (16.8)$$

The principal real logarithm is used in $(n+a)^{-s}$.

Proposition 16.3 (Conditional convergence). *The series (16.8) converges locally uniformly in $\Re s > 0$, and hence defines a holomorphic function there.*

Proof. The partial sums $S(N) = \sum_{n < N} \varepsilon_n$ are bounded by one. On a compact set $\Re s \geq \sigma > 0$, the first difference of $(n+a)^{-s}$ is uniformly $\mathcal{O}(n^{-\sigma-1})$. Summation by parts therefore gives uniform convergence of the tails. $\square$

Splitting even and odd indices gives an exact parameter equation.

Theorem 16.4 (Dyadic Dirichlet equation). *For $a > 0$ and $\Re s > 0$,*

$$D(s, a) = 2^{-s} \left(D\left(s, \frac{a}{2}\right) - D\left(s, \frac{a+1}{2}\right) \right). \quad (16.9)$$

Proof. Use $\varepsilon_{2n} = \varepsilon_n$ and $\varepsilon_{2n+1} = -\varepsilon_n$:

$$\begin{aligned} D(s, a) &= \sum_{n \geq 0} \frac{\varepsilon_n}{(2n + a)^s} - \sum_{n \geq 0} \frac{\varepsilon_n}{(2n + 1 + a)^s} \\ &= 2^{-s} \left(D\left(s, \frac{a}{2}\right) - D\left(s, \frac{a+1}{2}\right) \right). \end{aligned}$$

□

16.3 Mellin representation and entire continuation

Theorem 16.5 (Mellin integral). *For $a > 0$, initially when $\Re s > 1$,*

$$D(s, a) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-at} E(e^{-t}) dt. \quad (16.10)$$

The integral on the right is an entire function of s . Therefore (16.10) continues $D(s, a)$ to an entire function of $s \in \mathbb{C}$.

Proof. For $\Re s > 1$, absolute convergence permits termwise use of

$$(n + a)^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-(n+a)t} dt.$$

Summing $\varepsilon_n e^{-nt}$ gives $E(e^{-t})$.

At infinity, $E(e^{-t}) = 1 + \mathcal{O}(e^{-t})$, and the factor e^{-at} gives exponential decay. At zero, [theorem 16.1](#) gives decay faster than every positive power of t . Thus the integral and all of its s -derivatives converge uniformly on compact subsets of the entire s -plane. Multiplication by the entire function $1/\Gamma(s)$ preserves entire-ness. Equality on $\Re s > 1$, together with [theorem 16.3](#), gives the continuation by the identity theorem. □

Corollary 16.6 (Zeros at nonpositive integers). *For every $a > 0$ and every integer $r \geq 0$,*

$$D(-r, a) = 0. \quad (16.11)$$

Proof. The Mellin integral in (16.10) is finite at $s = -r$, while $1/\Gamma(s)$ has a zero there. □

These zeros are the analytic continuation of the finite Prouhet phenomenon: negative integer values formally ask for signed polynomial moments, and the flat Mellin kernel makes all of them vanish after continuation.

16.4 The harmonic kernel

At $s = 1$, the Mellin integral can be moved to the unit interval.

Corollary 16.7 (Dirichlet integral). *For $a > 0$,*

$$H(a) := D(1, a) = \sum_{n \geq 0} \frac{\varepsilon_n}{n + a} \quad (16.12)$$

$$= \int_0^1 x^{a-1} E(x) dx = \int_0^1 x^{a-1} \prod_{j \geq 0} (1 - x^{2^j}) dx > 0. \quad (16.13)$$

It satisfies

$$H(a) = \frac{1}{2} H\left(\frac{a}{2}\right) - \frac{1}{2} H\left(\frac{a+1}{2}\right). \quad (16.14)$$

Proof. Set $s = 1$ in (16.10) and substitute $x = e^{-t}$. The integrand is positive on $(0, 1)$. Equation (16.14) is (16.9) at $s = 1$. □

17 Infinite products with Thue–Morse exponents

17.1 Pair reduction

Suppose $R(n) \neq 0$ and the products converge. The even–odd recurrence gives the general reduction

$$\prod_{n \geq 0} R(n)^{\varepsilon_n} = \prod_{q \geq 0} \left(\frac{R(2q)}{R(2q+1)} \right)^{\varepsilon_q}. \quad (17.1)$$

A factor merely equal to $1 + \mathcal{O}(1/n)$ is often converted into a paired factor $1 + \mathcal{O}(1/n^2)$, explaining the good convergence of balanced rational products.

17.2 A two-parameter master product

For $a, b > 0$, define

$$G(a, b) = \prod_{n=0}^{\infty} \left(\frac{n+a}{n+b} \right)^{\varepsilon_n}. \quad (17.2)$$

The logarithm converges by summation by parts: the Thue–Morse partial sums are bounded and the first difference of $\log((n+a)/(n+b))$ is $\mathcal{O}(n^{-2})$.

Theorem 17.1 (Cocycle and dyadic functional equation). *For positive parameters,*

$$G(a, a) = 1, \quad G(a, b)G(b, c) = G(a, c), \quad G(b, a) = G(a, b)^{-1}, \quad (17.3)$$

$$G(a, b) = \left[\frac{G(a/2, b/2)}{G((a+1)/2, (b+1)/2)} \right]. \quad (17.4)$$

Proof. The cocycle identities are termwise. Pair indices $2n$ and $2n+1$:

$$\begin{aligned} & \left(\frac{2n+a}{2n+b} \right)^{\varepsilon_n} \left(\frac{2n+1+a}{2n+1+b} \right)^{-\varepsilon_n} \\ &= \left[\frac{n+a/2}{n+b/2} \frac{n+(b+1)/2}{n+(a+1)/2} \right]^{\varepsilon_n}. \end{aligned}$$

Multiply over n . □

The Dirichlet continuation identifies this product as a zeta-regularized derivative difference without any extra regularization being required.

Corollary 17.2 (Dirichlet derivative bridge). *For $a, b > 0$,*

$$\log G(a, b) = D'(0, b) - D'(0, a). \quad (17.5)$$

Proof. The difference $D(s, a) - D(s, b)$ can be differentiated termwise at $s = 0$, because the resulting log-ratio series converges by summation by parts. Its derivative is

$$\sum_{n \geq 0} \varepsilon_n \log \frac{n+b}{n+a} = -\log G(a, b).$$

Rearrange. □

17.3 The Woods–Robbins product

Theorem 17.3 (Woods–Robbins).

$$\prod_{n=0}^{\infty} \left(\frac{2n+1}{2n+2} \right)^{\varepsilon_n} = \frac{1}{\sqrt{2}}. \quad (17.6)$$

Proof. Let

$$A = \prod_{n \geq 0} \left(\frac{2n+1}{2n+2} \right)^{\varepsilon_n}, \quad B = \prod_{n \geq 1} \left(\frac{2n}{2n+1} \right)^{\varepsilon_n}. \quad (17.7)$$

Then

$$AB = \frac{1}{2} \prod_{n \geq 1} \left(\frac{n}{n+1} \right)^{\varepsilon_n}. \quad (17.8)$$

Split the product on the right into odd and even n . The odd part is A^{-1} because $\varepsilon_{2j+1} = -\varepsilon_j$, and the even part is B because $\varepsilon_{2j} = \varepsilon_j$. Hence

$$AB = \frac{1}{2} A^{-1} B. \quad (17.9)$$

Cancel B , obtaining $A^2 = 1/2$; positivity gives the stated root. $\square$

A second classical evaluation is

$$\prod_{n=0}^{\infty} \left(\frac{4n+1}{4n+3} \right)^{\varepsilon_n} = \frac{1}{2}. \quad (17.10)$$

It is the specialization $G(1/4, 3/4)$ and follows from the broader Thue–Morse gamma-product calculus of Allouche, Riasat, and Shallit [16].

17.4 Three dyadic block-product limits

Dyadic truncation respects exact block recursion. The following limits are convenient fingerprints of the same product theory:

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} \left(k + \frac{1}{2} \right)^{\varepsilon_k} = \frac{1}{2}, \quad (17.11)$$

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} (k+1)^{\varepsilon_k} = \frac{1}{\sqrt{2}}, \quad (17.12)$$

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} (k+1)^{(-1)^k \varepsilon_k} = \frac{1}{2\sqrt{2}}. \quad (17.13)$$

Pairing adjacent factors reduces the first two to $G(1/4, 3/4)$ and $G(1/2, 1)$, respectively; the third factors into their product. The evaluations are thus (17.10) and (17.6).

18 Base- q digit characters and generalized Prouhet identities

Let $q \geq 2$, let $s_q(n)$ be the sum of the base- q digits of n , and let $\zeta \neq 1$ satisfy $\zeta^q = 1$. Define

$$u_n = \zeta^{s_q(n)}. \quad (18.1)$$

Theorem 18.1 (Generalized digit product). *For $0 \leq r < q$,*

$$u_{qn+r} = \zeta^r u_n. \quad (18.2)$$

For every $m \geq 0$,

$$\sum_{n=0}^{q^m-1} u_n z^n = \prod_{j=0}^{m-1} \left(\sum_{r=0}^{q-1} \zeta^r z^{r q^j} \right). \quad (18.3)$$

Proof. Appending the base- q digit r increases the digit sum by r . In the product, choose one digit independently at each position; the exponent is the corresponding base- q integer and the coefficient is ζ raised to its digit sum. $\square$

Each factor vanishes at $z = 1$, since $1 + \zeta + \dots + \zeta^{q-1} = 0$.

Theorem 18.2 (Generalized Prouhet cancellation and sharp moment). *If p is a polynomial of degree less than m , then*

$$\sum_{n=0}^{q^m-1} \zeta^{s_q(n)} p(n) = 0. \quad (18.4)$$

At the first surviving degree,

$$\sum_{n=0}^{q^m-1} \zeta^{s_q(n)} n^m = m! \frac{q^{m(m+1)/2}}{(\zeta - 1)^m}. \quad (18.5)$$

Proof. Set $z = e^t$ in (18.3). Every factor has a zero at $t = 0$, proving cancellation below degree m . Its derivative at zero is

$$q^j \sum_{r=0}^{q-1} r \zeta^r = q^j \frac{q}{\zeta - 1} = \frac{q^{j+1}}{\zeta - 1}, \quad (18.6)$$

where the finite geometric sum was differentiated. Multiplying the m linear terms yields $q^{1+2+\dots+m}/(\zeta - 1)^m$; coefficient extraction contributes $m!$. $\square$

For $q = 2$ and $\zeta = -1$, the sequence is $u_n = \varepsilon_n$ and (18.5) becomes $(-1)^m m! 2^{\binom{m}{2}}$, exactly (9.18). The binary theory is therefore the first member of a root-of-unity digit-character family, while the particularly rich valuation, carry, and Pascal-row formulas remain genuinely base-two features.

19 Formula selection, computational checks, and a Lean roadmap

19.1 Formalized modules (August 2026)

The first tranches of the formalization now exist as nine compiled Lean modules in `Analysis/FabiusFunction`, complementing the corpus modules (`Parity.lean`, `TwoAdic.lean`, `ThueMorsePrefix.lean`, `ThueMorseGenerating.lean`, `ThueMorseExponential.lean`, `ThueMorseBinomialLog.lean`) that already covered the recursion, XOR, reflection, Lucas/Glaisher, binomial–logarithm, modulo-three formula, generating products, Prouhet moments, and prefix theory:

Module	Contents
<code>ThueMorseBooleanCube.lean</code>	Powerset reindexing kernel, digit dictionary, two-parameter master product over any commutative semiring, signed product over any commutative ring, weight enumerator.
<code>ThueMorseValuation.lean</code>	Successor law, weight subadditivity, ruler product, Catalan valuation, additive Kummer, borrow cocycle.
<code>ThueMorseDigits.lean</code>	Additive Legendre, floor sums, floor-parity formula, ‘testBit’ bridge, digit product.
<code>ThueMorseEnumerators.lean</code>	Weight distribution, evil/odious enumerations and composition laws, XOR autocorrelation, signed Hamming spheres.
<code>ThueMorseSparseProuhet.lean</code>	Arbitrary-weight Prouhet annihilation and sharp moment (theorem 9.4), Thue–Morse specializations, bit supports, sparse submask identities.
<code>ThueMorseRarefied.lean</code>	Modulo-three rarefaction recursion and closed vectors, Newman positivity at dyadic endpoints, block balance, even-modulus collapse.
<code>ThueMorseAperiodicity.lean</code>	Block-flip identities and non-eventual-periodicity.
<code>ThueMorseBlockAlgebra.lean</code>	Telescope and multiplicity ladder, degree and leading data of P_m , reciprocity (reverse and evaluation forms), cyclotomic multiplicity table, binomial-basis vanishing and sharp moment over any commutative ring.
<code>ThueMorseMoments.lean</code>	Hockey-stick column sum, complete binomial moment composition (theorem 9.12), reflection functional equation, midpoint parity selection rule, sharp centered moment, complete power-moment composition formula.

Wherever the Lean statement is stronger than the prose – arbitrary weights, arbitrary coefficient rings, hypothesis-free autocorrelation – the text above records the stronger form. The module proposals in the table below remain the plan for the layers not yet reached.

19.2 Which representation is best?

The formulas are equivalent mathematically but not proof-theoretically or computationally. The following table records the most economical interface for common tasks.

Goal	Recommended representation	Reason
Executable evaluation	Recurrence or machine popcount	Linear in the bit length; no large intermediate integers.
Dyadic reindexing	Block concatenation (2.6)	Quotient and remainder separate without carries.
Bitwise algebra	XOR character (2.9)	Makes ε a genuine group character.
Ordinary addition	Carry cocycle (4.10)	Isolates the exact obstruction to XOR multiplicativity.
Number-theoretic disguise	Central binomial valuation (4.4)	One classical integer stores the full binary weight.
Pascal-row interpretation	Glaisher count or modulo-three residue	Uses the odd support of one row; the residue form avoids logarithms.
Higher-dimensional Pascal	Multinomial count (5.18)	Works with any number of variables.
Formal power series	Finite/infinite products	Coefficients, Mahler equations, and zeros become immediate.

Goal	Recommended representation	Reason
Nonlocal coefficient formulas	Euler/Bell/determinant forms	Eliminate explicit digit notation and expose ruler convolutions.
Polynomial cancellation	Mixed finite differences	Proves all Prouhet moments in one operator identity.
Arbitrary Pascal row	Sparse signed product (10.2)	Restricts finite differences to the odd row support.
Finite harmonic analysis	Walsh character	The entire block is one pure frequency.
Ordinary spectral analysis	DFT sine product and Riesz density	Exposes roots of unity, amplitudes, and correlations.
Analytic continuation	Mellin integral (16.10)	Boundary flatness removes every Mellin singularity.
Infinite products	Master product $G(a, b)$	Pair reduction and cocycle laws organize evaluations.
Fabius approximation	Iterated-prefix generating function	Coefficient grids arise from $E(z)/(1-z)^k$.

19.3 A compact verification script

The following pseudocode is not a proof, but it is useful for regression tests when translating the atlas into Lean, a CAS, or a library implementation.

```

for n from 0 to N:
  s := popcount(n)
  eps := 1 - 2*(s mod 2)

  assert v2(factorial(n)) == n - s
  assert v2(binomial(2*n, n)) == s

  oddPascal := count k in [0, n] with binomial(n, k) odd
  assert oddPascal == 2^s
  assert (oddPascal mod 3) - 1 == (s mod 2)

  sparseParity := sum binomial(n, 2^j) over all 2^j <= n
  assert (-1)^sparseParity == eps

for m from 0 to M:
  P := sum_{n=0}^{2^m-1} eps(n)*z^n
  assert P == product_{j=0}^{m-1} (1-z^(2^j))

  for r from 0 to m-1:
    assert sum_{n=0}^{2^m-1} eps(n)*n^r == 0

  assert sum_{n=0}^{2^m-1} eps(n)*n^m
    == (-1)^m * factorial(m) * 2^(m*(m-1)/2)

```

19.4 A dependency-aware formalization order

A conservative Lean implementation should formalize the finite algebraic spine before introducing analysis.

1. **Binary layer.** Prove recurrence, block concatenation, XOR, reflection, and the two-kernel from a chosen binary-weight primitive.
2. **Valuation layer.** Establish the base-two Legendre formula, central-binomial valuation, Catalan valuation, and Kummer carry identity.
3. **Pascal/Frobenius layer.** Formalize Lucas support, Glaisher count, sparse probes, the modulo-three consequence, multinomial support, and Boolean Möbius inversion.
4. **Polynomial layer.** Define P_m , prove the finite product, reciprocity, cyclotomic multiplicities, and signed Pascal-support products over a suitable coefficient ring.
5. **Difference and moment layer.** Derive mixed finite differences, Prouhet cancellation, sharp moments, composition formulas, and sparse moments. Most statements can be done over $\mathbb{Q}$ before any topology is imported.
6. **Formal-series layer.** Add the Mahler equations, Euler transform, ruler convolution, Bell/determinant forms, and the Artin–Schreier equation over $\mathbb{F}_2[[z]]$.
7. **Analytic layer.** Only then introduce compact-uniform convergence, Fourier evaluation, Riesz measures, Mellin transforms, and infinite products.

Possible module boundaries are

Proposed module	Main declarations
<code>ThueMorseValuation.lean</code>	Factorial, central-binomial, Catalan, successor, and carry formulae.
<code>ThueMorsePascal.lean</code>	Lucas submasks, Glaisher count, sparse probes, modulo-three formula.
<code>ThueMorseMultinomial.lean</code>	Frobenius factorization and $O_r(n) = r^{\text{wt}_2(n)}$.
<code>ThueMorseBooleanMobius.lean</code>	Submask incidence algebra and inverse parity matrix.
<code>ThueMorseGeneratingProduct.lean</code>	P_m , cyclotomic multiplicities, Mahler equations.
<code>ThueMorseProuhetMoments.lean</code>	Difference operators, all moments, centered parity.
<code>ThueMorseSparseProuhet.lean</code>	Arbitrary Pascal-support products and sparse moments.
<code>ThueMorseFormalSeriesF2.lean</code>	Artin–Schreier equation and explicit solution.
<code>ThueMorseFourier.lean</code>	Finite DFT, Walsh transform, product-to-sum identities.
<code>ThueMorseDirichlet.lean</code>	Boundary flatness, Mellin integral, entire continuation.
<code>ThueMorseRarefied.lean</code>	Root-of-unity filters, modulo-three closed forms, odd-modulus matrix recursion.
<code>ThueMorseWords.lean</code>	Uniform recurrence, aperiodicity, and the elementary complexity bounds.

The last module is analytically the heaviest. The first seven are finite or formal and can deliver useful lemmas to the existing Fabius development without waiting for measure theory or complex Mellin infrastructure.

19.5 Proof-dependency graph

The central dependencies can be summarized as

$$\begin{aligned}
 \text{binary digits} &\implies \begin{cases} \text{recursion and XOR,} \\ \text{Lucas/Frobenius support,} \\ \text{finite product } P_m, \end{cases} \\
 P_m &\implies \begin{cases} \text{Mahler and Euler equations,} \\ \text{Fourier/Walsh products,} \\ \text{order-}m \text{ zero at 1,} \end{cases} \\
 \text{order-}m \text{ zero} &\implies \begin{cases} \text{Prouhet cancellation,} \\ \text{all finite moments,} \\ \text{iterated-prefix flatness,} \\ \text{Riesz-product normalization.} \end{cases}
 \end{aligned} \tag{19.1}$$

Legendre/Kummer form a parallel arithmetic branch; Lucas and the signed support product join that branch to the finite-difference branch at sparse Prouhet identities. Boundary flatness of the infinite product then opens the Mellin/Dirichlet branch.

20 Worked examples and consistency checks

20.1 One index through many representations

Take $n = 13 = (1101)_2$. The occupied positions are 0, 2, 3, so

$$\text{wt}_2(13) = 3, \quad \tau(13) = 1, \quad \varepsilon_{13} = -1. \tag{20.1}$$

The floor, factorial, and central-binomial routes give

$$\sum_{j \geq 1} \left\lfloor \frac{13}{2^j} \right\rfloor = 6 + 3 + 1 = 10, \tag{20.2}$$

$$(-1)^{13 - \nu_2(13!)} = (-1)^3 = -1, \tag{20.3}$$

$$\nu_2 \binom{26}{13} = 3. \tag{20.4}$$

The distinguished Pascal entries are

$$\binom{13}{1} = 13, \quad \binom{13}{2} = 78, \quad \binom{13}{4} = 715, \quad \binom{13}{8} = 1287. \tag{20.5}$$

Their parities are 1, 0, 1, 1, exactly the four low bits of 13, and their sum 2093 is odd. Row 13 has

$$\nu(13) = 2^3 = 8 \tag{20.6}$$

odd entries, so $\nu(13) \bmod 3 = 2$ and the logarithm-free formula gives $\tau(13) = 2 - 1 = 1$. In three variables, exactly

$$O_3(13) = 3^3 = 27 \tag{20.7}$$

coefficients of $(x + y + z)^{13}$ are odd.

The submasks are

$$0, 1, 4, 5, 8, 9, 12, 13, \tag{20.8}$$

and their signed support polynomial is

$$R_{13}(z) = (1 - z)(1 - z^4)(1 - z^8). \quad (20.9)$$

Its derivatives of orders 0, 1, 2 at one vanish, and

$$\sum_{k \preccurlyeq_2 13} \varepsilon_k k^3 = -3! 2^{0+2+3} = -192. \quad (20.10)$$

All these evaluations recover the same three occupied bits through different algebraic interfaces.

20.2 An all-order dyadic moment

For $m = 3$ and $r = 5$, direct summation gives

$$M_{3,5} = \sum_{n=0}^7 \varepsilon_n n^5 = -6720. \quad (20.11)$$

The positive compositions of 5 into three parts are

$$(1, 1, 3), (1, 2, 2), (1, 3, 1), (2, 1, 2), (2, 2, 1), (3, 1, 1).$$

Formula (9.20) gives

$$M_{3,5} = -5! \sum_{\substack{p_0+p_1+p_2=5 \\ p_j \geq 1}} \frac{2^{p_1+2p_2}}{p_0!p_1!p_2!} = -6720. \quad (20.12)$$

No recursion through lower moments is needed.

20.3 A Fourier sample

For $m = 4$, every even DFT frequency vanishes. At $k = 1$,

$$\widehat{\varepsilon}_4(1) = (2i)^4 e^{-15\pi i/16} \sin \frac{\pi}{2} \sin \frac{\pi}{4} \sin \frac{\pi}{8} \sin \frac{\pi}{16}. \quad (20.13)$$

The factorization exposes both the exact phase and the hierarchy of four dyadic scales, which direct summation of sixteen complex terms obscures.

20.4 Autocorrelation from the recursion

Starting with $\eta(0) = 1$, the fixed-point equation at one gives $\eta(1) = -1/3$. Then

$$\begin{aligned} \eta(2) &= \eta(1) = -\frac{1}{3}, \\ \eta(3) &= -\frac{\eta(1) + \eta(2)}{2} = \frac{1}{3}, \\ \eta(5) &= -\frac{\eta(2) + \eta(3)}{2} = 0. \end{aligned}$$

These are simultaneously combinatorial averages and Fourier coefficients of the singular continuous Thue–Morse measure.

20.5 A complete four-bit block

At $m = 4$,

$$\begin{aligned} P_4(z) &= 1 - z - z^2 + z^3 - z^4 + z^5 + z^6 - z^7 \\ &\quad - z^8 + z^9 + z^{10} - z^{11} + z^{12} - z^{13} - z^{14} + z^{15} \\ &= (1 - z)(1 - z^2)(1 - z^4)(1 - z^8). \end{aligned} \quad (20.14)$$

The factor $(1 - z)^4$ divides the product, so

$$\sum_{n < 16} \varepsilon_n n^r = 0 \quad (0 \leq r < 4), \quad \sum_{n < 16} \varepsilon_n n^4 = 4! 2^{\binom{4}{2}} = 1536, \quad (20.15)$$

the sign being positive because $(-1)^4 = 1$. The residue-class sums modulo three are, by [theorem 15.1](#) at level $m = 4$,

$$(A_{4,0}^{(3)}, A_{4,1}^{(3)}, A_{4,2}^{(3)}) = (6, -3, -3). \quad (20.16)$$

Their total is zero, as block balance requires, yet the distribution among residue classes is already strongly skewed toward the multiples of three.

20.6 A carry-corrected addition

Take $a = 13 = (1101)_2$ and $b = 7 = (0111)_2$. Then $\varepsilon_{13} = \varepsilon_7 = -1$, while $a + b = 20 = (10100)_2$ has $\varepsilon_{20} = +1$. Kummer’s formula [\(4.9\)](#) counts

$$\nu_2 \binom{20}{13} = \text{wt}_2(13) + \text{wt}_2(7) - \text{wt}_2(20) = 3 + 3 - 2 = 4 \quad (20.17)$$

carries, and the correction $(-1)^4 = 1$ in [\(4.10\)](#) restores multiplicativity: $\varepsilon_{20} = (-1)(-1)(+1)$. The example shows why a yes/no record of “was there a carry” cannot suffice: the sign needs the carry count with propagation, taken modulo two.

21 Concluding synthesis

The formula in Section 11.8 of the Fabius–Rvachev exposition is not an isolated trick. It lies at the intersection of four structures:

binary digits $\longleftrightarrow$ Frobenius/Pascal support $\longleftrightarrow$ Boolean-lattice characters $\longleftrightarrow$ dyadic finite differences
--

(21.1)

The signed Thue–Morse sequence is the common coefficient system of this translation dictionary.

The arithmetic formulas hide digit parity in factorial valuations, central binomial coefficients, Catalan numbers, and carry counts. The Pascal formulas recover it either globally, from the number of odd row entries, or sparsely, from the power-of-two columns. The modulo-three identity shows that the exact base-two logarithm is optional: the exponent parity can also be read from the cycle of powers of two in a finite residue field. Multinomial support turns the same mechanism into a whole $r^{\text{wt}_2(n)}$ family.

The product $P_m(z) = \prod_{j < m} (1 - z^{2^j})$ is the central unifying engine. At $z = 1$ it has an exact high-order zero and produces Prouhet cancellation. At $z = e^t$ it produces every finite moment. At roots of unity it produces the dyadic DFT. On the Boolean cube it is a pure Walsh character. After taking absolute squares it produces Riesz densities. After division by $(1 - z)^k$ it produces iterated-prefix grids and their terminal flatness. After the limit $m \rightarrow \infty$, it becomes a Mahler function, an Euler transform of the ruler sequence, and the Mellin kernel of an entire Dirichlet family.

The added Mellin theory sharpens the analytic picture: the infinite product is not merely small near one, but flat beyond every algebraic order, with a quadratic leading term in $\log(1/t)$. That single estimate explains why all potential Mellin poles disappear and why the shifted Thue–Morse Dirichlet series has zeros at every nonpositive integer. Its derivative at zero then recovers the logarithms of the gamma-like products generated by Thue–Morse exponents.

For the Fabius project, the most useful lesson is proof-interface diversity. No one formula dominates every task. Block multiplicativity is ideal for dyadic reindexing; the carry cocycle for ordinary sums; Pascal support for characteristic-two arguments; finite differences for cancellation; Bell polynomials for all moments; Walsh characters for finite harmonic analysis; and the Mellin product for analytic continuation. Formalizing several of these interfaces would not duplicate mathematics: it would let later Fabius proofs choose the representation adapted to the operation being performed.

A The first 64 terms

The rows display $\tau(n)$; replacing 0 by +1 and 1 by −1 gives ε_n .

Range	successive values							
0–7	0	1	1	0	1	0	0	1
8–15	1	0	0	1	0	1	1	0
16–23	1	0	0	1	0	1	1	0
24–31	0	1	1	0	1	0	0	1
32–39	1	0	0	1	0	1	1	0
40–47	0	1	1	0	1	0	0	1
48–55	0	1	1	0	1	0	0	1
56–63	1	0	0	1	0	1	1	0

Every row of length eight is a copy or complement according to the sign of its block index, as predicted by (2.6).

B A compact identity dictionary

For quick reference, the principal exact translations are

$$\begin{aligned}
 \tau(n) &\longleftrightarrow \varepsilon_n = 1 - 2\tau(n), \\
 \varepsilon_n &\longleftrightarrow (-1)^{\text{wt}_2(n)}, \\
 \text{wt}_2(n) &\longleftrightarrow n - \nu_2(n!), \\
 \text{wt}_2(n) &\longleftrightarrow \nu_2\binom{2n}{n}, \\
 b_j(n) &\longleftrightarrow \binom{n}{2^j} \bmod 2, \\
 2^{\text{wt}_2(n)} &\longleftrightarrow \#\{k : \binom{n}{k} \text{ odd}\}, \\
 \varepsilon_n &\longleftrightarrow [z^n] \prod_{j \geq 0} (1 - z^{2^j}), \\
 \sum_{n < 2^m} \varepsilon_n f(x + n) &\longleftrightarrow (-1)^m \Delta_1 \Delta_2 \cdots \Delta_{2^{m-1}} f(x),
 \end{aligned}$$

$$P_m(e^{-2\pi i k/2^m}) \longleftrightarrow \widehat{e}_m(k),$$

$$E(e^{-t}) \longleftrightarrow \Gamma(s)D(s, a) \text{ under Mellin transformation.}$$

Every arrow denotes an exact identity or invertible conversion, not an asymptotic analogy.

C Draft-to-section concordance

The four source drafts overlap substantially. The following table records the main distinctive material retained from each, together with additions made in the unified treatment.

Source	Distinctive contributions represented in this atlas
Draft 1	Carry and XOR viewpoints; Boolean Möbius inversion and inverse Pascal parity matrix; ruler Euler transform; partition, Bell, and Hessenberg determinant formulae; finite-difference positivity and analytic transforms.
Draft 2	Sparse power-of-two Pascal columns; all-order dyadic moments; automaton/tensor/Walsh forms; exact DFT and Riesz products; master infinite products; base- q root-of-unity extension.
Draft 3	Multinomial support and exact r -nomial logarithms; sparse Prouhet theorem on arbitrary Pascal rows; autocorrelation recurrences; Artin–Schreier series over $\mathbb{F}_2$; Catalan and fractional-part formulae.
Draft 4	Logarithm-free modulo-three Pascal formula; sinc-product interval signs; evil/odious enumerators; integer algebraic lift; trigonometric product-to-sum identities; explicit iterated-prefix/Fabius bridge.
Unified additions	Dyadic weight-distribution chapter; centered-moment parity selection; all sparse Pascal-support moments; elementary boundary asymptotic for $E(e^{-t})$; entire shifted Dirichlet family, its nonpositive-integer zeros, and its derivative bridge to exponent products; sharp first moment for the general base- q character.
Parallel atlas 1	Self-contained natural-boundary proof; supertile arguments for uniform recurrence and the linear complexity bound $p(\ell) < 8\ell$; the literature panorama of arithmetic subsequences (primes, squares, cubes, level of distribution).
Parallel atlas 2	Gelfond’s sharp uniform exponent with the lower bound at cube roots of unity; the odd-modulus matrix recursion; the autocorrelation Mahler equation; subtraction/borrow cocycle; weighted digit enumerator; Hankel/Padé framing of irrationality measure.
Parallel atlas 3	Exact modulo-three rarefaction vectors with the two-scale constants; Newman positivity framing; complement/reversal closure of the factor language; critical exponent statement; Mahler value transcendence and $\mu = 2$ exposition; four-bit block and carry-corrected worked examples.
Second-pass additions	Even-modulus collapse of rarefied sums from exact prefix identities; the characteristic-zero/characteristic-two reading of the lacunary digit-sum series (6.19); Morse–Hedlund lower complexity bound paired with the elementary upper bound.

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